L-2/T-1/EEE **Date: 07/09/2025**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2023-2024

Sub: **EEE 211** (Continuous Signals and Linear Systems)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer to **Question No. 1 is compulsory**.

Answer any **TWO** from questions 2-4.

All the symbols have their usual meanings. Assume reasonable values for any missing data.

1. (a) Consider the signal shown in Fig. for Q. No. 1(a). **(11)** (CO1)

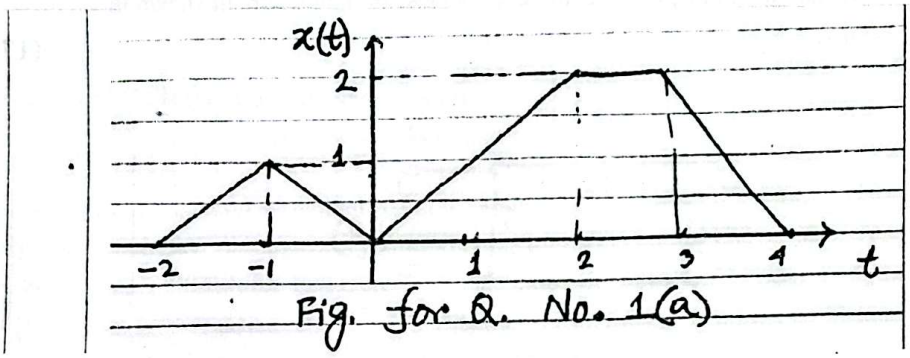


Fig. for Q. No. 1(a)

(i) Draw $x(-3t+2)$, $x'(t)$ and $x''(t)$.

(ii) Express $x'(t)$ using elementary signals, e.g. $\delta(t)$, $u(t)$, $r(t)$ etc.

(b) Consider a system with input $x(t)$ and output $y(t)$. If the system is described by the following differential equation, then determine whether the system is (i) linear; (ii) time invariant. **(7)** (CO2)

$$\frac{dy(t)}{dt} + t^2 y(t) = (2t+3)x(t).$$

(c) Consider the system shown in Fig. for Q. No. 1(c). Given, **(17)** (CO2)

$$h_1(t) = e^{-2t}u(t),\ h_2(t) = e^{-2t}u(t),\ h_3(t) = e^{-t}u(t),\ h_4(t) = \delta(t),\ h_5(t) = e^{-3t}u(t)$$

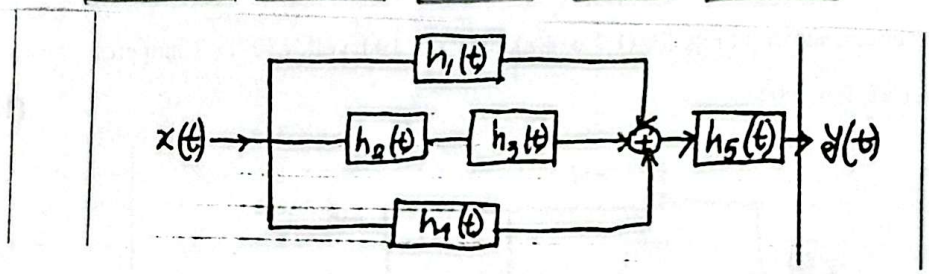


Fig. for Q. No. 1(c)

Using time domain technique,

(i) Determine whether the system is BIBO stable.

(ii) Also, determine the output $y(t)$ for the input $x(t) = e^{-t}u(t)$.

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2. (a) Graphically determine the convolution of the pair of signals in Fig. for Q. No. 2(a). **(18)**

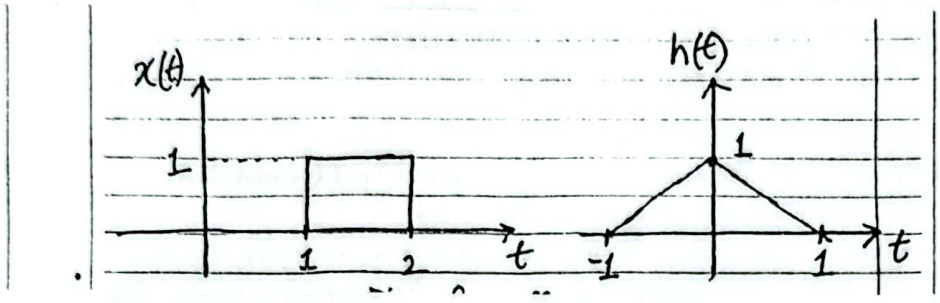


Fig. for Q. No. 2(a)

(b) Using time domain technique, find the impulse response of the LC circuit shown in Fig. for Q. No. 2(b). **(17)**

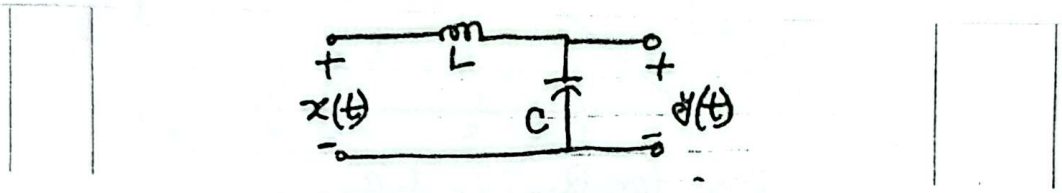


Fig. for Q. No. 2(b)

3. (a) Derive the first canonical state space representation of the system described by the differential equation as shown below. **(18)**

$$y'''(t)-2y''(t)+y'(t)+4y(t)=x'''(t)+5x(t)$$

Also, determine the stability of the system from the state space representation.

(b) Consider the system described by **(17)**

$$v'(t)=\begin{bmatrix}-3 & 4\\ -2 & 3\end{bmatrix}v(t)+\begin{bmatrix}1\\ 3\end{bmatrix}x(t), \text{ and}$$

$$y(t)=[-1 \quad -1]v(t)+2x(t)$$

Find the impulse response, $h(t)$ of the system [use Cayley Hamilton theorem].

4. (a) For the circuit shown in Fig. for Q. No. 4(a), let $v_C(0^-)=1$ volt, $i_L(0^-)=2$ amperes and $x(t)=u(t)$, find $y(t)$. **(18)**

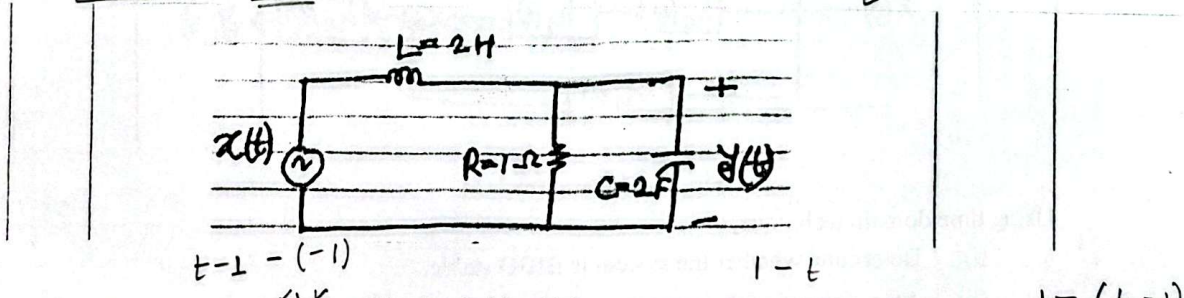


Fig. for Q. No. 4(a)

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Contd.... for Q. No. 3

(b) Draw the force-voltage analogous electrical circuit and write the equation of motion in terms of the given mechanical quantities of the system shown in Fig. for Q. No. 4(b). **(17)**

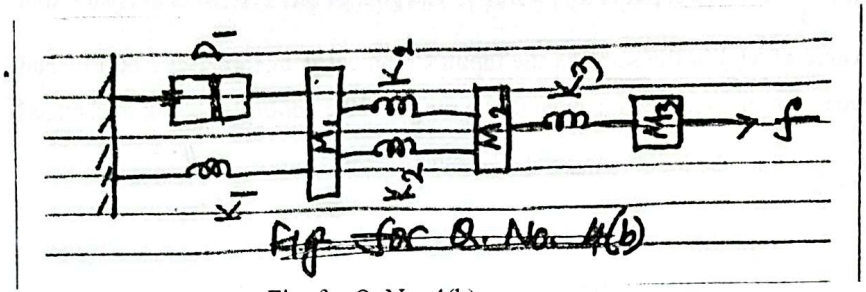


Fig. for Q. No. 4(b)

SECTION – B

There are **FOUR** questions in this section. Answer to **Question No. 5 is compulsory.** Answer any **TWO** from questions 6-8.

5. (a) In communication system, different techniques are used to transmit signal from one place to another through the same channel or medium. One such technique is quadrature multiplexing. In quadrature multiplexing, two carriers are used in the same frequency band but they are 90° out of phase. Such a system is presented in Fig. for Q. 5(a). Assume, $\omega_c = 4\omega_m$. **(20)** (CO3)

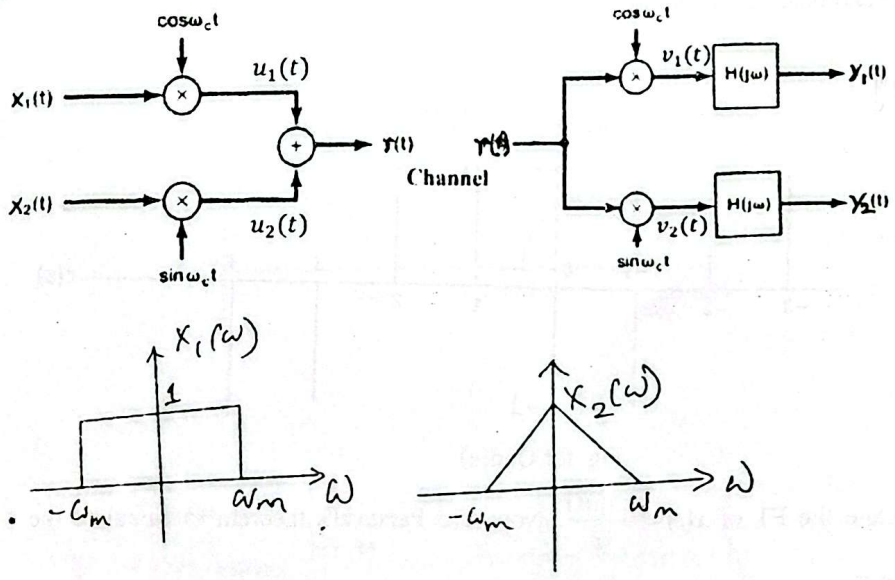


Fig. for Q. 5(a)

(i) Design the filters $H(j\omega)$ such that, $y_1(t) = x_1(t)$ and $y_2(t) = x_2(t)$.

(ii) Draw the magnitude spectrum at different point of this system, namely, $u_1(t), u_2(t), v_1(t), v_2(t)$, and $y_2(t)$.

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Contd.... for Q. No. 5

(b) The system in Fig. for Q. 5(b) shows a plant $H(s)\frac{1}{s+2}$ controlled by a controller block $H_c(s)\frac{s+1}{s}$ and the input is $x(t)=Au(t)$. The goal of this system is to ensure that the final value of $y(t)$ is the same as the input. Comment if this goal can be fulfilled for the given system. What will happen to the output if the controller block is changed to $H_c(s)\frac{1}{s-2}$ while the input remains the same? **(15)**
(CO4)

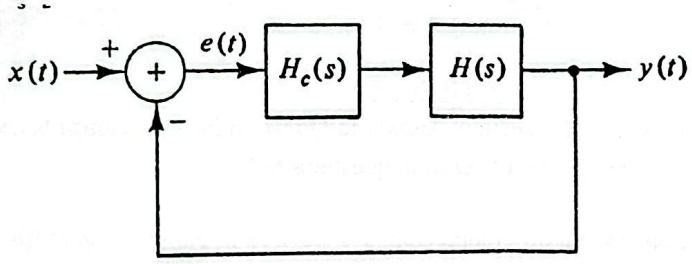


Fig. for Q. 5(b)

6. (a) Determine the Fourier Transform of signal shown in Fig. for Q. 6(a) and plot the amplitude and phase spectrum of the. **(15)**

corresponding

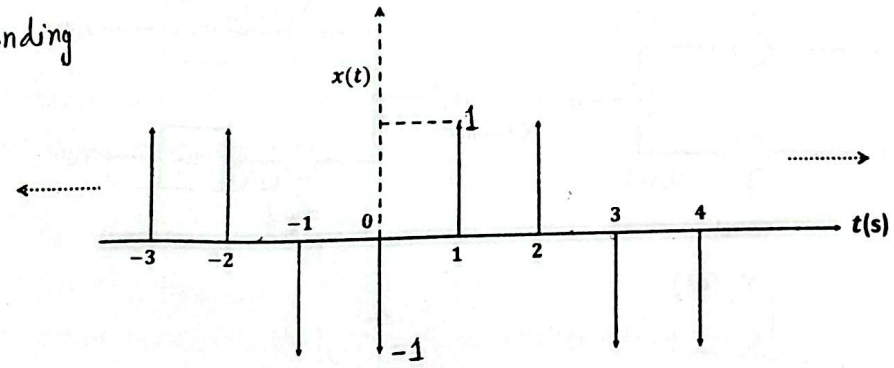


Fig. for Q. 6(a)

(b) Calculate the FT of $x(t)=\frac{\sin(at)}{\pi t}$. Now, use Parseval's theorem to calculate the energy of this signal. **(10)**

(c) Derive the Fourier Transform of the unit step function, $u(t)$. Using this result, prove the integration property of Fourier Transform. **(10)**

7. (a) Show that the three functions shown in Fig. for Q. 7(a) are pairwise orthogonal over the interval (−1, 2). Calculate the value of A that will make these signals orthonormal to each other. **(10)**

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Contd.... for Q. No. 7(a)

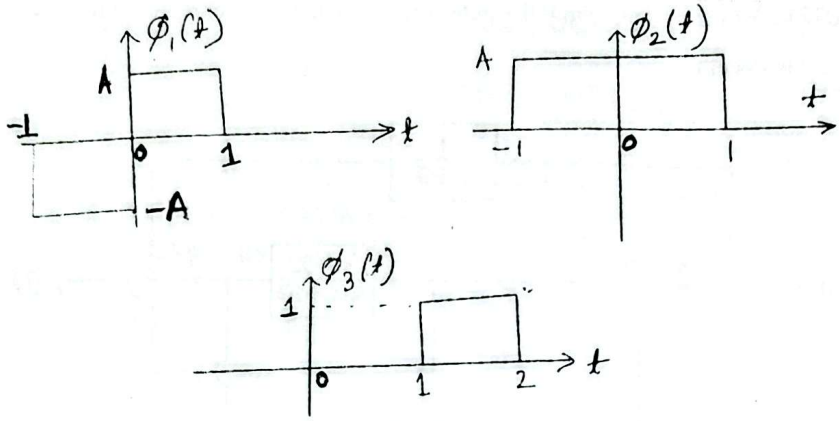


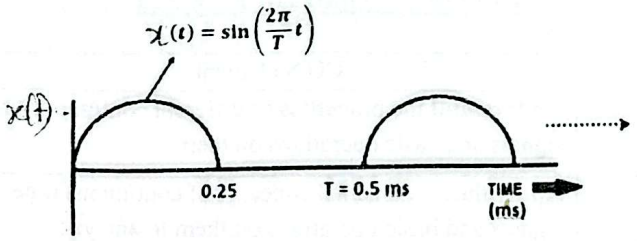
Fig. for Q. 7(a)

(b) The signal x(t) shown in Fig. for Q. 7(b)(i) is used as input to two different LTI systems shown in Fig. for Q. 7(b)(ii). System 1 is a differentiator, and the frequency response of system 2 is shown in Fig. for Q. 7(b)(iii). The cutoff frequency (f_c) of the lowpass filter shown in Fig. for Q. 7(b)(iii) is 1 kHz. **(25)**

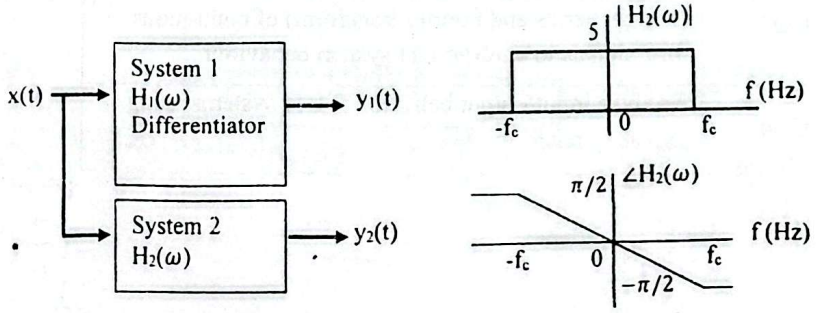
(i) Find the Fourier Series coefficients of $y_1(t)$ by using the Fourier Series coefficients of $x(t)$.

(ii) Find the time domain expression of $y_2(t)$. Mention which harmonics will have nonzero values.

(iii) Compare how each system behaves toward the low-frequency and high-frequency components of the input.



(i)



(ii) (iii)

Fig. for Q. 7(b)

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8. (a) Find the system transfer function for the system shown in the Fig. for Q. 8(a). Also, determine poles of the system, show the poles on the s-plane and comment on system stability of this system. **(17)**

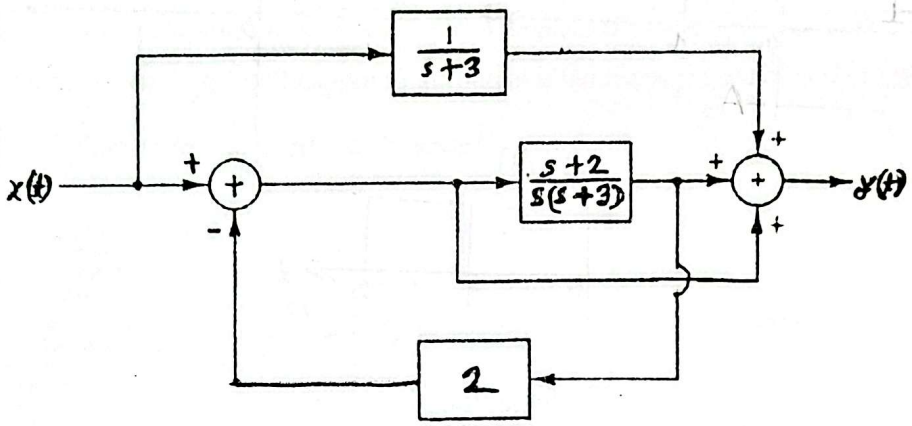


Fig. for Q. 8(a)

(b) Find the output $y(t)$ of the system represented by the following differential equation: **(18)**

$\frac{d^2y(t)}{dt^2}+5\frac{dy(t)}{dt}+6y(t)=\frac{dx(t)}{dt}+x(t)$ by analyzing the system in Laplace domain. Here, $y(0-)=2, y'(0-)=1$ and the input is $x(t)=e^{-4t}u(t)$. Identify the zero-input response $y_{zi}(t)$ and the zero-state response $y_{zs}(t)$.

Course Outcomes of EEE 211

CO No.	CO Statement
CO1	**Understand** the properties of different continuous time signals and basic operations on them
CO2	**Apply** the fundamental concepts of continuous time signals and basic operations on them to analyze input/output behavior of LTI systems in time-domain.
CO3	**Apply** the concept of frequency domain transformation (Fourier series and Fourier transform) of continuous time signals to analyze LTI system behaviour
CO4	**Analyze** input/output behavior of LTI systems using Laplace transform

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The figures in the margin indicate full marks. Symbols bear their usual meanings

USE SEPARATE SCRIPTS FOR EACH SECTION

The Course Outcomes (COs) of each part of Question 1 and 5 are mentioned on the right most column. The COs of the Course are mentioned at the end of the question paper.

SECTION – A

There are **FOUR** questions in this section. Answer to Question no 1 is compulsory. Answer any 2 questions from questions 2-4.

1. (a) Consider the design of a thermostat system for a room in a building, where the input $x(t)$ is the desired room temperature and the output $y(t)$ is the actual room temperature. For the purpose of this design, the system will be modeled as an LTI system with a system function $H(s)$. Determine which of the functions $H_1(s)$ or $H_2(s)$ given below would be a more desirable choice for $H(s)$. **(18)** (CO4)

$$H_1(s) = \frac{s^2}{s^2+9} \qquad H_2(s) = 1 + \frac{1}{20}\left[\frac{s(s+10)}{(s+10)^2+1}\right]$$

What percentage of incremental change of temperature will be achieved in 1.5s for your chosen system?

(b) Derive the $(f-i)$ electrical equivalent of the mechanical system shown in Fig. for Q.1(b), which shows a rotary two-disc pulley, a horizontally translating body of mass m_3 with a spring and a damper attached to it, and two rigid massless wires (the vertical one has a spring connected to it). The pulley is formed of two concentric disks with masses m_1 and m_2, and radius $r1$ and $r2$, respectively, that form a solid piece. Assuming that the pulley and the translator body undergo small movements, write down all associated motion equations. **(17)** (CO5)

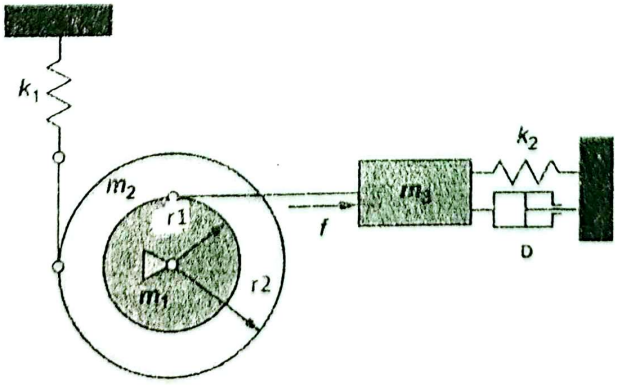


Fig for Q.1(b)

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2. (a) Determine the closed-form expression of Fourier transform $X(\omega)$ of $x(t)$, which is shown in Fig. for Q.2(a). **(10)**

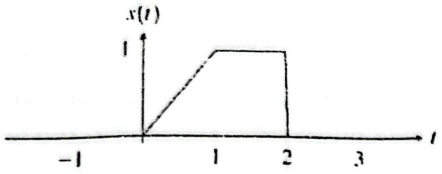


Fig for Q.2(a)

(b) An amplitude-modulated signal z(t) is generated using the system given in Fig. for Q. 2(b) where x(t) is down sampled version of m(t). The spectrum of the input signal m(t) and the frequency response of the filter $H_f(\omega)$ are also shown in the figure. If the frequency related parameters of the system are such that $\omega_m = \omega_f = \frac{\omega_0}{2}$ **(25)**

(i) Draw the spectrums of x(t), y(t) and z(t).

(ii) Evaluate the energy of the modulated signal z(t) in terms of system parameters

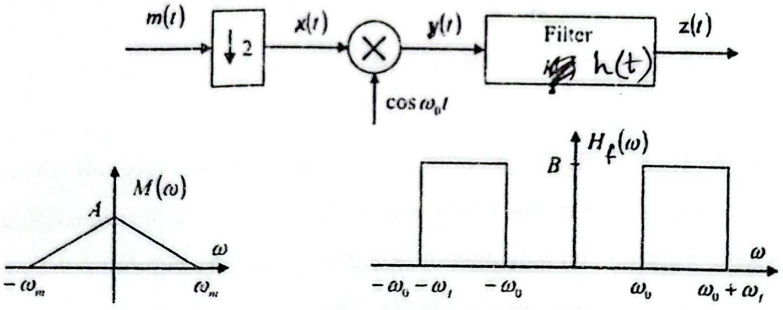


Fig for Q.2(b)

3. (a) Consider the circuit shown in Fig. for Q. 3(a), in which the source voltage is given as **(18)**

$$v(t) = \begin{cases} 2\,V & t < 0 \\ 4e^{-2t}\,V & t \geq 0 \end{cases}$$

Determine the $i_R(t)$ for $t \geq 0$ using Laplace transform.

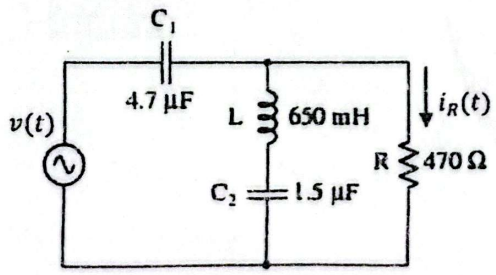


Fig for Q.3(a)

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Contd... Q. No. 3

(b) For an LTI system with input $x(t)$ and output $y(t)$, the state space parameters are **A**, **B**, **c**, and d. The state space representation is as follows: **(17)**

$$v'(t) = \mathbf{A}v(t) + \mathbf{b}x(t)$$

$$y(t) = \mathbf{c}v(t) + dx(t)$$

Determine the expression of system transfer function $H(s)$, and the impulse response $h(t)$ using Laplace transform where the parameters are

$$\mathbf{A} = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \mathbf{c} = [-1 \quad -1] \quad d = 1$$

What is the ROC of unit step response of $Y(s)$?

4. (a) In an AM communication channel, a band-limited signal $x(t)$ having a bandwidth of $[-\omega_m, \omega_m]$ is modulated by a carrier signal $\cos\omega_c t$, and the modulated signal is transmitted through an ideal channel. The frequency of the local oscillator at the receiver end is $\cos(\omega_c t - \theta)$. If the receiver end ideal low-pass filter has a gain of 2 and a cut-off frequency of ω_f $(\omega_M \leq \omega_f \leq 2\omega_c - \omega_M)$, then what is the expression of output signal? **(15)**

(b) An LTI system has a transfer function given by **(20)**

$$H(s) = \frac{s^2 + s + 1}{s^3 + 4s^2 + 3s + 2}$$

(i) Draw the simulation diagram of the system in the second canonical form

(ii) Write the expression of the corresponding state equation

5 compulsory.

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) A periodic signal $x(t)$ is represented by a trigonometric Fourier series- **(10)**

$x(t) = 2 + 3\cos 2t + 4\sin 2t + 2\sin(3t + 30°) - \cos(7t + 150°)$ **[CO3]**

Express this series as a compact trigonometric Fourier series, i.e., in the form $x(t) = A_0 + \sum_n A_n \cos\left(\frac{2\pi nt}{T} + \phi_n\right)$. Sketch the amplitude spectra with proper x-and Y-axis labels.

(b) The input $x(t)$ and output y(t) of an LTI system are showing in Fig. for Q. 5(b). Find the expressions of the outputs of this system for the following two inputs: **(15)**

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Contd... Q. No. 5(b)

(i) $x\left(t-\frac{1}{2}\right)-x\left(t+\frac{1}{2}\right)$; (ii) $\frac{dx(t)}{dt}$. Sketch these two outputs with proper x-and y-axes labels. **[CO2]**

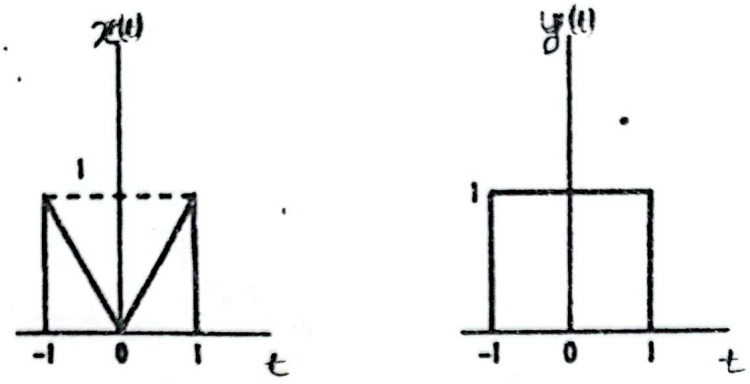


Fig. for Q. 5(b)

(c) For the signal $x(t)$ shown in Fig. for Q. 5(c), write the signal as a linear combination of elementary signals. Sketch the signal $\frac{dx(t)}{dt}$ with proper labels on the x-and y-axes. **(10)**

[CO1]

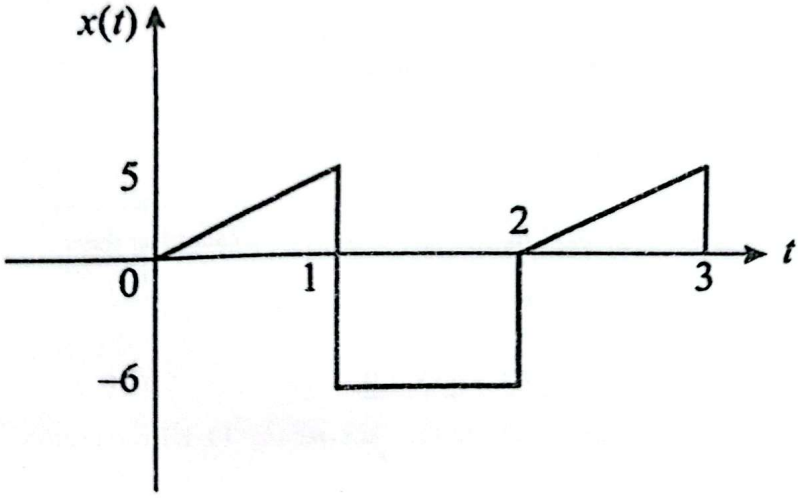


Fig. for Q. 5(c)

6. (a) Check whether the following signals can be used as a mathematical model of a delta function: **(20)**

(i) $p_1(t)=\lim_{\epsilon\to 0}\left(\epsilon\pi\cosh\frac{t}{\epsilon}\right)^{-1}$.

(ii) $p_2(t)=\lim_{\epsilon\to 0}\frac{1}{\pi}\frac{\sin\epsilon t}{t}$.

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~~Contd... Q. No. 6(a)~~

(b) Consider the system described by the following state equations **(15)**

$$v'(t) = Av(t) + bx(t)$$

$$y(t) = cv(t) + dx(t)$$

If can be shown that, by performing a simple change of variable $\mathbf{z}(t) = \mathbf{P}v(t)$, the new state equation can be written as

$$z'(t) = A_1 z(t) + b_1 x(t)$$

$$y(t) = c_1 z(t) + d_1 x(t)$$

If **P** is a square matrix with det(**P**) > 0, then express $\mathbf{A}_1$, $\mathbf{b}_1$, $\mathbf{c}_1$, and d_1 in terms of **A**, **b**, **c**, d, and **P**.

7. (a) For the signal $x(t)$ shown in Fig. for Q. 7(a), find the expression of the exponential Fourier series. **(20)**

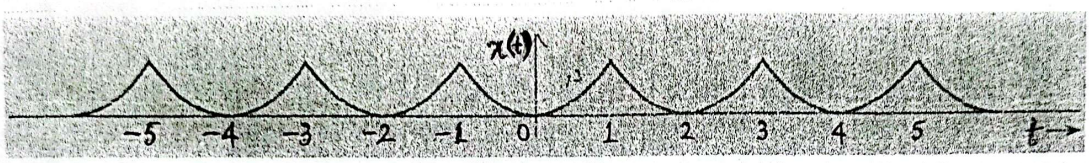


Fig. for Q. 7(a)

(b) Consider an LTI system where $x(t)$ = h(t) = t[u(t) - u(t-1)]. Find the expression of the output $y(t)$ of the system for all t. **(15)**

8. (a) Find the analytical expression of the trigonometric Fourier series (that is, in the form $\delta_{T_0}(t) = A_0 + \sum_n A_n \cos\left(\frac{2\pi nt}{T} + \phi\right)$) for the impulse train $\delta_{T_0}(t)$ depicted in Fig. for Q. 8(a). Sketch the amplitude spectrum with proper x- and y-axes labels. **(20)**

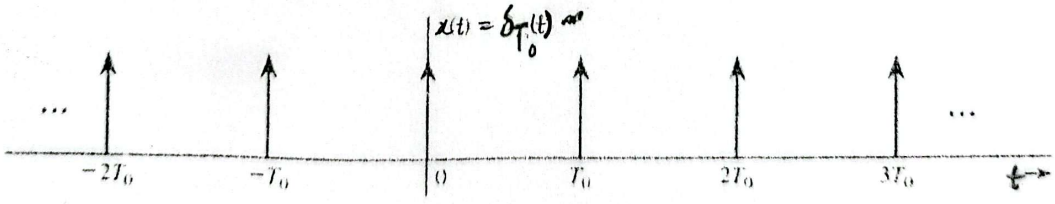


Fig. for Q. 8(a)

(b) Consider a system with input $x(t)$ and output $y(t)$. If the system is describe by the following differential equation, then comment on whether the system is (i) linear; (ii) time invariant. **(15)**

$$\frac{dy(t)}{dt} + t^2 y(t) = (2t+3)x(t)$$

L-2/T-1/EEE **Date: 28/03/2023**

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Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

The symbols have their usual meanings.

1. (a) Is the following signal periodic? If so, what is the period? **(10)**

$$x(t)=\sqrt{e^{j5\pi t}}+\sin^2\left(3t-\frac{\pi}{4}\right)$$

(b) Determine whether the signal $e^{-3t}\sin\left(\frac{2\pi}{3}t-4\right)u(t)$ is a power or an energy signal.

Justify your answer. **(10)**

(c) Give a signal $x(t)=\delta(t-1)+u(t)u(t-3)+u(\cos t)$ **(15)**

Compute $y(t)=\int_{-\alpha}^{t}x(\tau)d\tau$ and sketch $y(1/2)$.

2. (a) The input $x(t)$ and the impulse response of an LTI system is shown in Fig. for Q. 2(a). Compute the output $y(t)$ using the graphical method. Also, write the analytical expression of $y(t)$. **(17)**

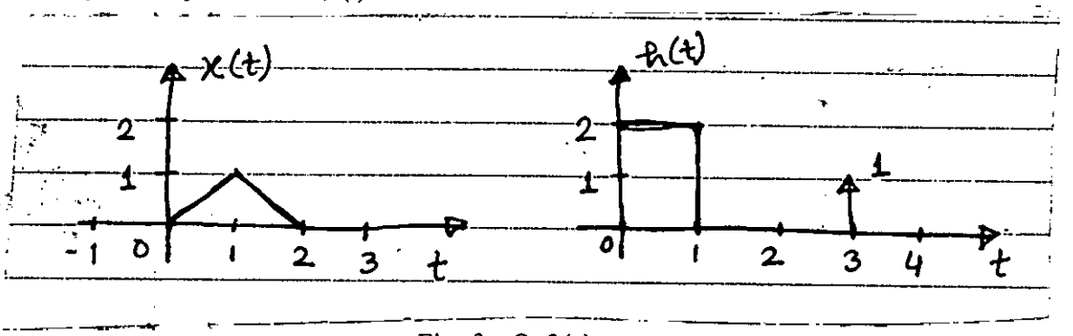
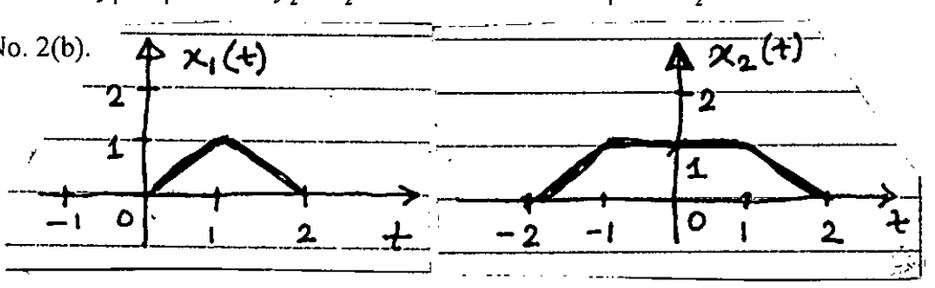


Fig. for Q. 2(a)

(b) Let h be the impulse response of an LTI system, and let x_1, x_2, y_1, y_2 denote functions such that $y_1 = x_1 * h$ and $y_2 = x_2 * h$. The functions x_1 and x_2 are shown in Fig. for Q. No. 2(b). **(18)**



Contd P/2

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3. (a) Let $x(t)$ be a periodic signal with a period of T = 4. The signal in one period can be written as $x_T(t) = \delta(t-1) + 2[u(t-2) - u(t-4)]$ $0 \le + \le 4$. The signal is applied to the circuit shown in Fig. for Q. 3(a). Plot the magnitude and phase spectra of input $x(t)$ and output $y(t)$. **(20)**

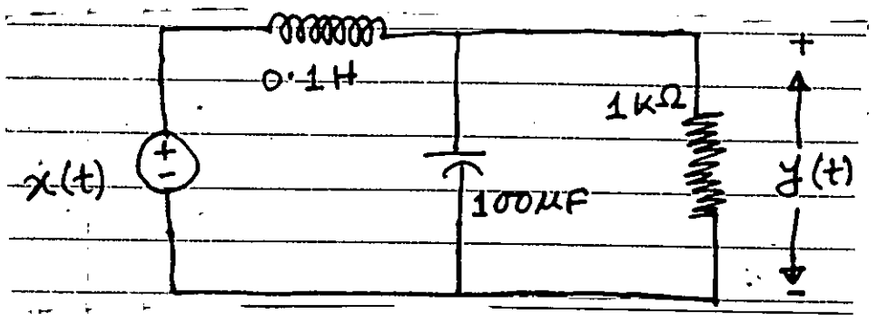


Fig. for Q. 3(a)

(b) Show that three functions as shown in Fig. for Q. 3(b) are pair-wise orthogonal over the interval (–2, 2). How can you make these functions as an orthonormal set? Express the function $x(t)$ in terms of the orthonormal set, where, **(15)**

$$x(t) = \begin{cases} 1. & \text{for} \quad 0 \le t \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Validate the Perseval's relation in this case.

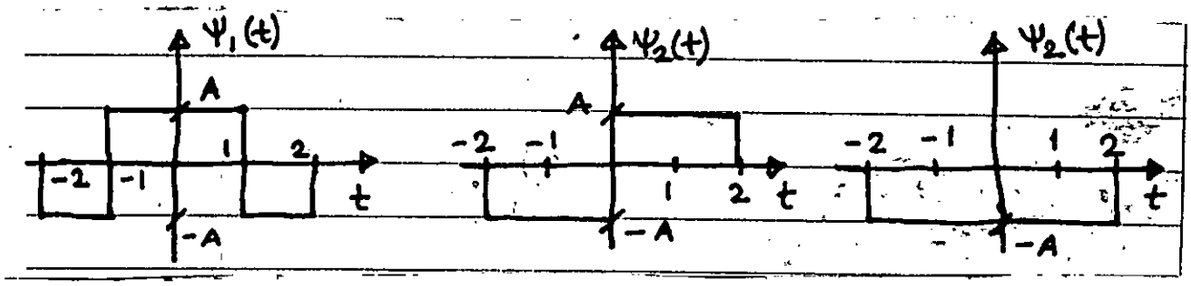


Fig. for Q. 3(b)

4. An LTI system is described by the following input-output relation: **(35)**

$$y''(t) + 7y'(t) - 12y(t) = x''(t) + 3x'(t) + 4x(t); \quad y'(0) = -2, \; y(0) = 0$$

(i) Draw the simulation diagram of the system in second canonical form.

(ii) Write the state equation of the system.

(iii) Find the state transition matrix, and hence evaluate the $h(t)$ of the system.

(iv) Comment on the stability of the system, and evaluate the unit-step response of the system.

(v) Evaluate the matrix that will transform the state equation of the second canonical form to the first canonical form.

You can use Laplace transform as and when required.

Contd P/3

EEE 211

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

5. (a) The Fourier series representation of a periodic signal $\tilde{x}(t)$ is given by **(18)**

$$\tilde{x}(t) = \sum_{n=-\infty}^{\infty} C_n e^{jnw_o t}$$

What does happen if the waveform, $\tilde{x}(t)$, is kept unchanged but the period T is increased? Show mathematically that in the limit as $T \to \infty$, we can write one period of $\tilde{x}(t)$ as

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(w) e^{jwt}\, dw$$

(b) Determine the time-domain signal $x(t)$ that has the Fourier transform $X(w)$, as shown in Fig. for Q. 5(b) **(17)**

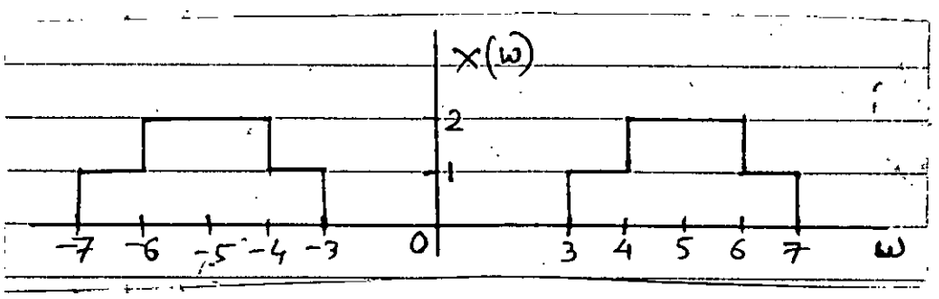


Fig. for Q. 5(b)

6. (a) An analog bandbass signal $x_a(t)$ is given as input to the system as shown in Fig. for Q. 6(a). **(20)**

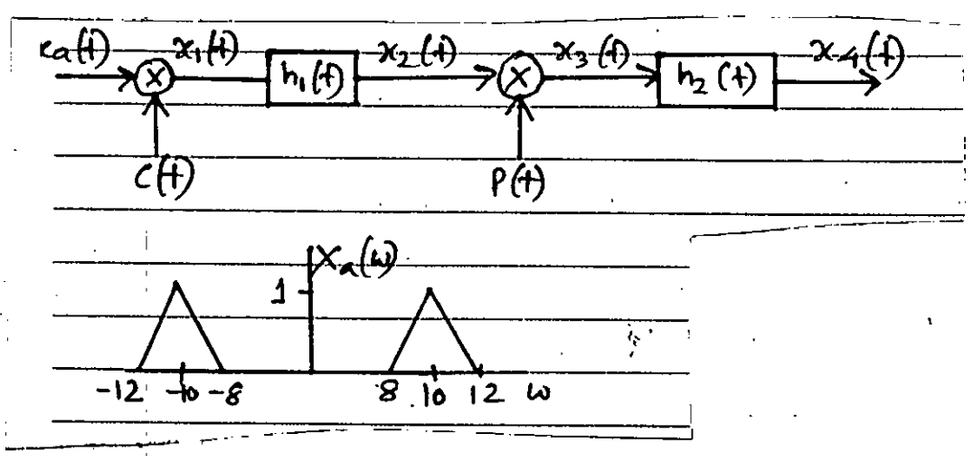


Fig. for Q. 6(a)

Contd P/4

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Contd.... for Q. 6(a)

Here, $c(t)=\cos(10t)$

$$h_1(t)=\frac{\sin(2t)}{\pi t}$$

$$p(t)=\sum_{n=-\infty}^{\infty}\delta\left(t-\frac{2\pi n}{5}\right)$$

$$h_2(t)=2\frac{\sin(2t)}{\pi t}\cos(10t)$$

Sketch the spectra of the signals $x_1(t), x_2(t), x_3(t)$ and $x_4(t)$.

(b) Consider a non-causal LTI system with the impulse response $h(t)=e^{2t}u(-t)$. **(15)**

(i) Determine the 3-dB bandwidth of the system,

(ii) Determine the duration of the impulse response, at which $h(t)$ drops to $\frac{1}{e}$ of its value at $t = 0$.

(iii) Determine the duration-bandwidth product of the system.

7. (a) Determine the Laplace transform and sketch the ROC of each of the following signals: **(12)**

(i) $e^{-at}u(t), a>0$

(ii) $e^{-at}u(t), a<0$, and

(iii) $-e^{-at}u(-t), a<0$.

(b) use convolute property to find the time-domain signal corresponding to the Laplace transform **(12)**

$$X(s)=\frac{1}{(s-a)^3}$$

(c) Solve the differential equation using the Laplace transform: **(11)**

$$y'(t)+2y(t)=e^{-3t}u(t),\quad y(0^-)=1$$

8. (a) Consider a control system shown in Fig. for Q. 8(a). here, the input to the system is the reference signal $r(t)$, and $w(t)$ is introduced to model any disturbance in the system. **(18)**

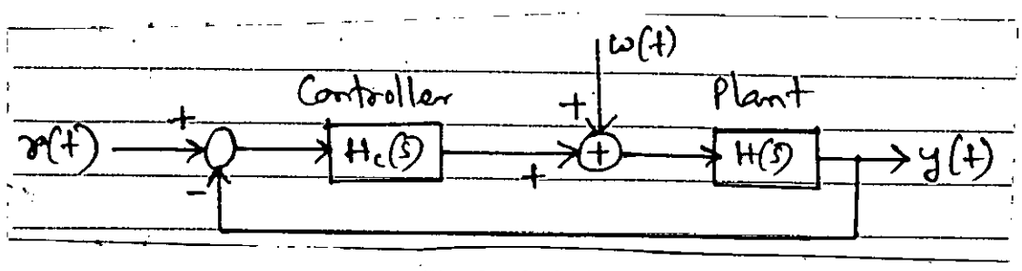


Fig. for Q. 8(a)

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Contd.... for Q. No. 8(a)

(i) Determine the transfer function of the system, $H_d(s) = \frac{Y(s)}{W(s)}$, with respect to the disturbance signal.

(ii) Show that the output of the system for a unit step-disturbance will be zero at steady-state condition, if the controller $H_c(s)$ has a zero at $s = 0$.

(b) Draw the force-voltage analogous electrical circuit and write the equation of motion in term of the given mechanical quantities of the system shown in Fig. for Q. 8(b). **(17)**

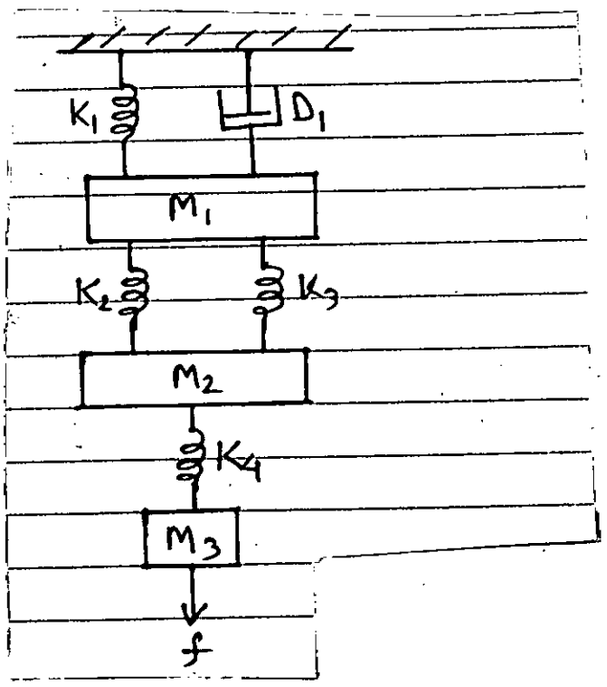


Fig. for Q. 8(b)

L-2/T-1/EEE

Date: 30/03/2022

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2020-2021

Sub: **EEE 211** (Continuous Signals and Linear Systems)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE** questions.

Symbols have their usual meanings.

1. (a) A signal $x(t)$ has a Fourier transform **(15)**

$$X(\omega)=\frac{j\omega}{\omega^2-3j\omega-2}$$

Evaluate the transform of $\frac{dx(t)}{dt}\cos(t)$.

(b) **(20)**

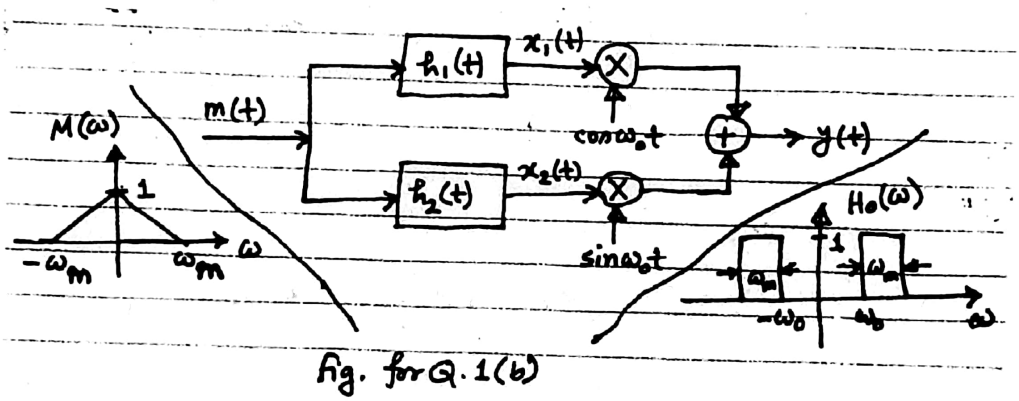


Fig. for Q. 1(b)

Consider the system shown in Fig. for Q.1(b). The impulse responses $h_1(t)$ and $h_2(t)$ have frequency responses as follows:

$$H_1(\omega)=\frac{1}{2}[H_0(\omega-\omega_0)+H_0(\omega+\omega_0)]$$

$$H_2(\omega)=\frac{j}{2}[H_0(\omega-\omega_0)-H_0(\omega+\omega_0)]$$

(i) Sketch the spectrum of $y(t)$,

(ii) Evaluate the energy of $y(t)$ using Persaval's relation.

Assume, $\omega_0 >> \omega_m$.

2. (a) Find the power spectral density of the signal **(8)**

$$x(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT).$$

(b) Evaluate the 3-dB bandwidth of the signal $x(t)=e^{-3t}\cdot u(t)$ **(7)**

(c) A causal LTI system is represented by the following differential equation **(20)**

$$\frac{d^2y(t)}{dt^2}+6\frac{dy(t)}{dt}+8y(t)=2x(t)$$

(i) Find the frequency response of the system.

(ii) Find the response of the system when $x(t)=te^{-2t}u(t)$.

Contd P/2

EEE 211

3. (a) Consider an LTI system shown in Fig. for Q. 3(a). The system functions are **(15)**

$$H_1(S)=\frac{1}{S^2+bS+(b-3)};\quad H_2(S)=-2$$

where b is a real constant. Find he transfer function of the overall system. Also determine the values of the parameter b so that the system will be stable.

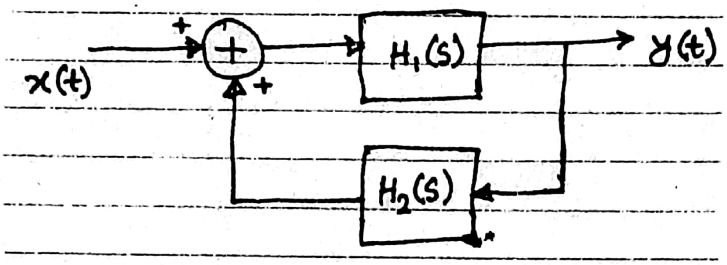


Fig. for Q. 3(a)

(b) **(20)**

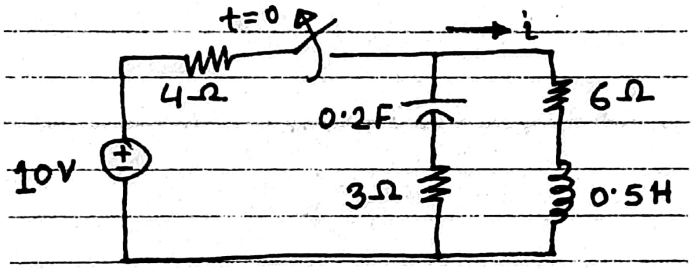
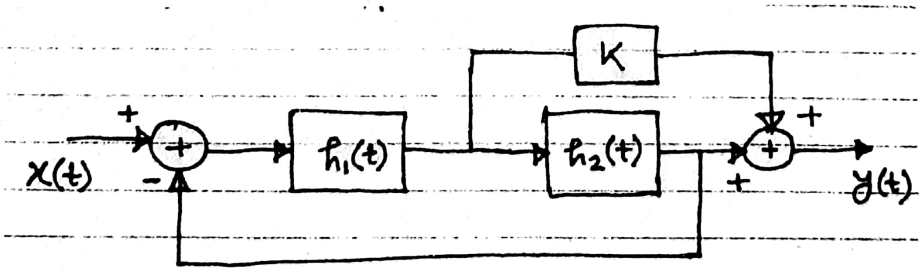


Fig. for Q. 3(b)

Using Laplace transform find $i(t), t>0$ in the circuit shown in Fig. for Q. 3(b). Assume that the circuit has reached steady state at $t = 0^-$.

4. (a) Find the transfer function and unit-step response of the overall system shown in Fig. for Q. 4(a). **(20)**



Here, $H_1(S)=\frac{1}{S+2}$, $H_2(S)=\frac{S}{S+3}$, and $k = 2$

Fig. for . Q. 4(a)

Contd P/3

EEE 211

Contd...Q. No. 4

(b) Draw the torque-current analogous electrical circuit and write the equation of rotational motion in terms of given mechanical quantities of the system shown in Fig. for Q. 4(b). **(15)**

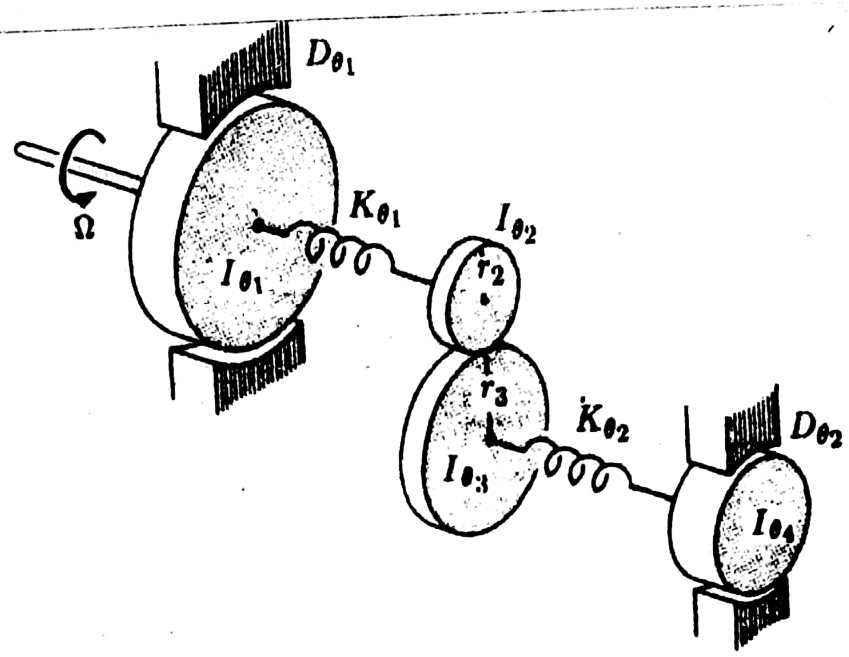


Fig. for Q. 4(b).

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE** questions.

Answer in brief and to the point.

Make reasonable assumptions on any missing information.

5. (a) Three signals $x_1(t)$, $x_2(t)$ and $x_3(t)$ are shown in Fig. for Q. No. 5(a).

(i) Express $x_1(t)$ and $x_2(t)$ in terms of ramp function $r(t)$ and unit step function $u(t)$. **(10)**

(ii) Draw $x_1(t+2)r(t-2)$ and $x_2(t-2)u(-2t)$. **(10)**

(iii) Express $x_3(t)$ in terms of $x_1(t)$. **(5)**

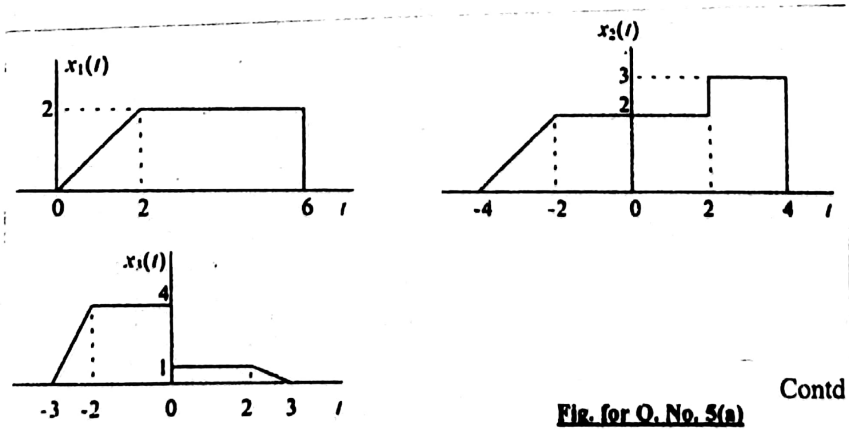


Fig. for Q. No. 5(a)

Contd P/4

EEE 211

Contd...Q.No.5

(b) Analytically determine whether the following signals are power or energy signal. **(10)**

(i) $u(t-6)-u(-t+8)$, and (ii) $2r(t-4)u(t)$.

6. (a) Consider a continuous time system with input-output relationship: $y(t)=tx(t+2)$. Determine whether the system is (i) linear, (ii) causal, (iii) time-variant, (iv) memory less and (v) stable. You will have to use time-domain techniques. **(15)**

(b) Using the graphical convolution computation, determine the output $y(t)$ of an LTI system if the input is $x(t)$ = rect(t/2) and the system impulse response is $h(t)=r(t)u(-t+1)$. Also, write the analytical expression of $y(t)$ and draw it. **(10)**

(c) Consider an RC series circuit as an LTI system with an input voltage $x(t)$. The voltage across the resistor $v_R(t)$ is the output of the circuit. Assume that the circuit is initially relaxed. Using time-domain analysis, determine the impulse response $h(t)$ of the circuit. **(10)**

7. (a) For the signals given in Fig. for Q. No. 7(a), first compute the exponential Fourier series coefficients of $x(t)$. Then, using the properties of Fourier series, determine the Fourier series coefficients of $y(t)$. Also draw the amplitude spectrums of $x(t)$ and $y(t)$. **(20)**

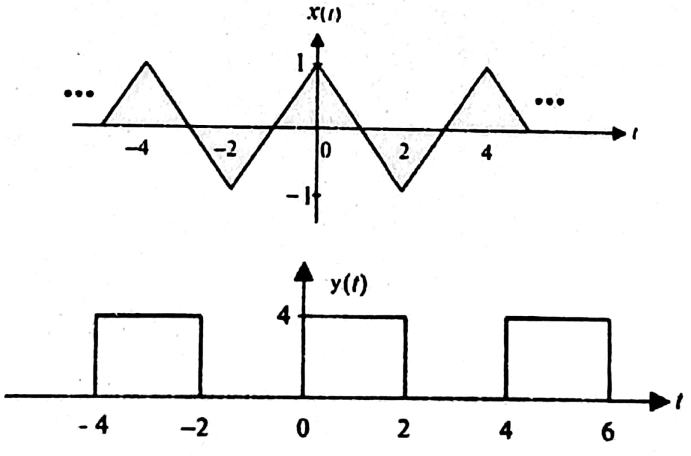


Fig. for Q. No. 7(a)

(b) An LTI system is modeled by the following differential equation: **(15)**

$$y''(t)+3y'(t)+2y(t)=x'(t)+x(t)$$

(i) Draw the amplitude spectrum and the phase spectrum of the output signal $y(t)$ if the input is given by $x(t)=2\cos(t)+6\cos(3t)$.

(ii) Calculate the average power of $y(t)$ using Parseval's theorem.

Contd P/5

EEE 211

8. (a) An LTI system is described by the following differential equation. Using state-variable technique, find the time-domain impulse response of the system. **(20)**

$$y''(t)+6y'(t)+8y(t)=x'(t)+x(t)$$

Use Second Canonical form to write the state equations and Cayley-Hamilton theorem for finding the impulse response. Assume that the system is initially relaxed.

Also determine whether the system is stable or not.

(b) For the LTI system given in Question 8(a), determine the *s*-domain expression of the state-transition matrix of the system. Also, derive the *s*-domain expression of unit step response of the system. **(15)**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B.Sc. Engineering Examination 2019-20

Sub: **EEE 211** (Continuous Signals and Linear Systems)

Full Marks: 210 Time 2:30 Hours

The Figures in the margin indicate full marks

USE SEPARATE SCRIPTS FOR EACH SECTION

There are 5 page(s) in this question paper.

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**

All the symbols have their usual meanings

Assume reasonable values for missing data.

Q1. A continuous-time system is given by (35)

$$y(t)=\frac{1}{3}\int_{-\infty}^{\infty} x(\tau)\left[\delta(t-\tau)+\delta(t-\tau+t_o)+\delta(t-\tau-t_o)\right]d\tau.$$

(i) Explain what this system does.

(ii) Is this system BIBO-stable? Justify your answer.

(iii) Is the system linear? Justify your answer.

(iv) Is the system memoryless? Justify your answer.

(v) Is the system causal? Justify your answer.

(vi) Is the system time invariant? Justify your answer.

Q2. Suppose an LTI system has impulse response *h(t)* and input *x(t)*. Fig. for Q2 shows *x(t)* and *h(t+1)*. (35)

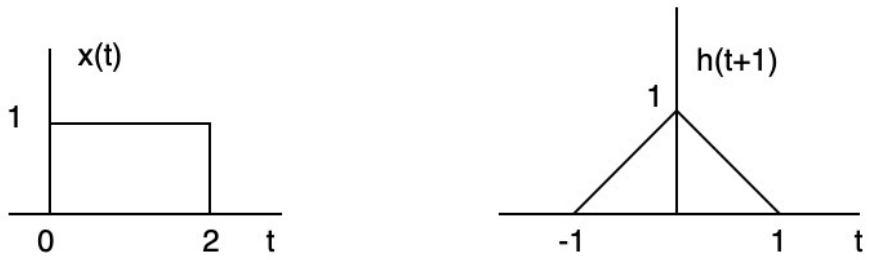


Fig. for Q2

(i) Is the system causal? Justify your answer.

(ii) Use the graphical convolution technique to determine $y(t) = x(t) * h(t)$.

(iii) Compare x(t) and y(t), and explain what this system does.

(iv) Sketch the output of the system for a different input $z(t) = x(t) - x(t-4)$.

Q3. The output of a multipath system y(t) may be expressed in terms of the input $x(t)$ as (35)

$$y(t) = x(t) + ax(t - T)$$

where a and T represent the relative strength and time delay of the second path.

(i) Find the transfer function, $H(s) = Y(s) / X(s)$, of the multipath system.

(ii) Determine the transfer function of the inverse system so that we can recover $x(t)$ from $y(t)$. Show that the inverse system can be realized by a feedback system.

(iii) Determine the impulse response of the inverse system. What condition must be satisfied for the inverse system to be both causal and stable?

Q4 Consider the mechanical system shown in Fig. for Q4 with input force, $f(t)$ and output displacement, $x(t)$. (35)

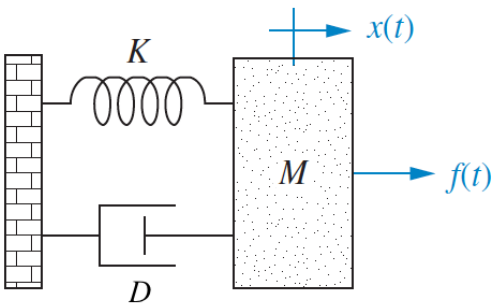


Fig for Q4

(i) Draw the analogous electrical circuit of the mechanical system considering

force-voltage analogy.

(ii) Determine the state variable model of the analogous electrical circuit considering current and charge on the capacitor as the state variables.

(iii) Determine the impulse response of the state variable model considering $M = 1$, $K = 40$ and $D = 12$.

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**

All the symbols have their usual meanings

Assume reasonable values for missing data.

Q5. (a) Describe the signal *x(t)* in Fig. for Q5(a) by a single expression valid for all *t*. (15)
Sketch the time-compressed version of *x(t)* by a factor of 3.

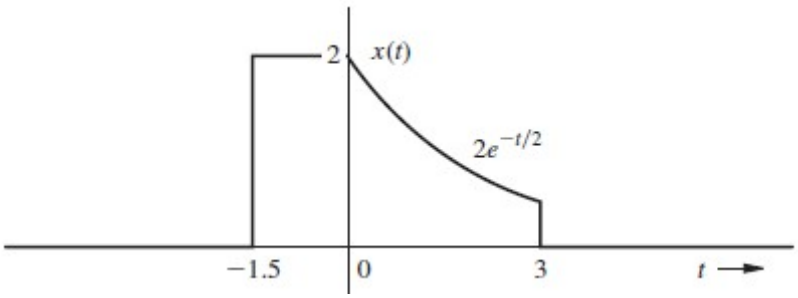


Fig. for Q5(a)

(b) A full-wave rectifier circuit shown in Fig. for Q5(b) is used to obtain a DC signal from a sinusoid *sin(t)*. The rectified signal *x(t)* is applied to the input of a lowpass RC filter. Find the system transfer function *H(ω)* and the filter output *y(t)*. (20)

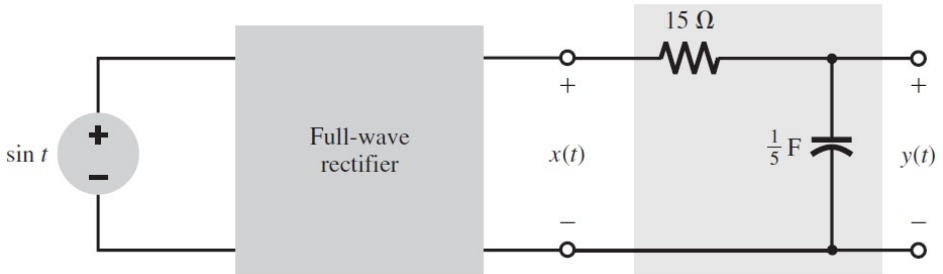


Fig. for Q. 5(b)

Q6. (a) If the duration of a signal *x(t)* is defined as the time at which *x(t)* drops to *1/e* of the value at *t = 0*, using this concept find the durations of the following signals: (15)

i) $x_1(t) = A \exp(-t/T)\, u(t)$;

ii) $x_2(t) = x_1(3t)$;

iii) $x_3(t) = x_1(t/2)$.

(b) State and prove the duality property of the Fourier transform. (20)

Q7. (a) Find the exponential Fourier series for the two impulse trains shown in Fig. for Q7(a). (15)

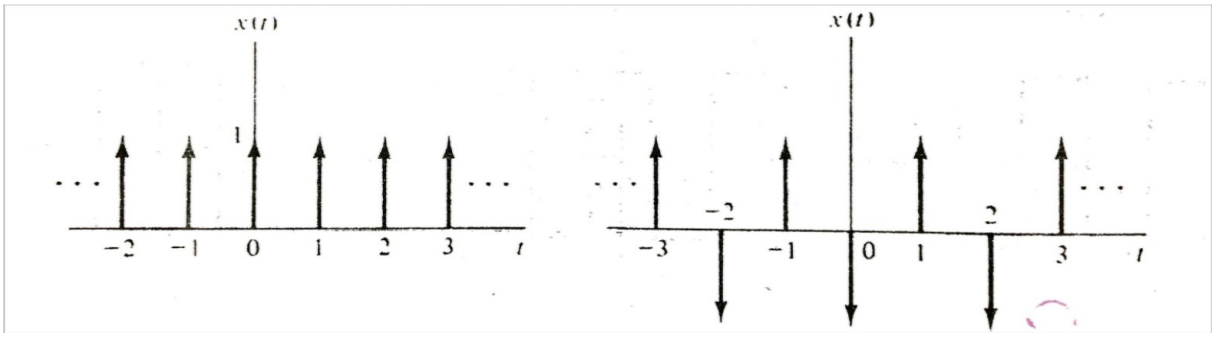


Fig. for Q7(a)

(b) For the system shown in Fig. for Q7(b), derive the expression for *y(t)* for (20)

$$x(t)=\frac{\sin\left(\frac{\omega_B t}{2}\right)}{\pi t},\ p(t)=\sum_{n=-\infty}^{\infty}\delta\left(t-\frac{n\pi}{\omega_B}\right),\ h(t)=\frac{\sin\left(\frac{3\omega_B t}{2}\right)}{\pi t}.$$

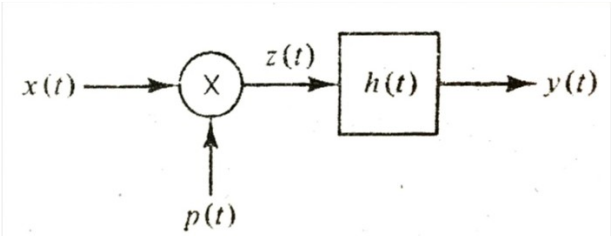


Fig. for Q7(b)

Q8. (a) With detailed calculations, show that the Fourier transform of a sequence of (15)

impulses in the time domain yields a sequence of impulses in the frequency domain.

(b) The trigonometric Fourier series of the periodic signal *x(t)* with period *T* can be written as $x(t)=a_0+\sum_{n=1}^{\infty}\left(a_n\cos\frac{2n\pi t}{T}+b_n\sin\frac{2n\pi t}{T}\right)$. For the triangular signal shown in Fig. for Q8(b), calculate the values of a_0, a_n and b_n. (20)

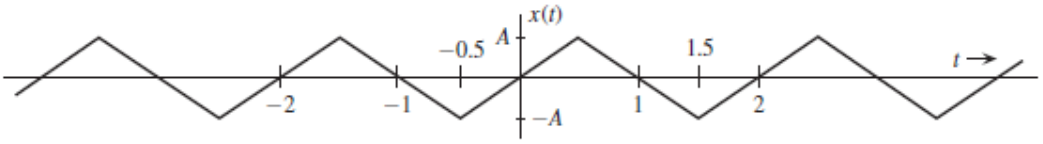


Fig. for Q8(b)

L-2/T-1/EEE **Date: 20/10/2019**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2018-2019

Sub: **EEE 211** (Continuous Signals and linear systems)

Full Marks: 210 Time: 3 Hours

USE SEPARATE SCRIPTS FOR EACH SECTION

The figures in the margin indicate full marks

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

1. (a) State and explain the orthogonal condition of a set of signals over an interval. Are the exponential functions exp ($j2t$) and exp ($j4t$) orthogonal over the interval t = [0,π] ? **(10)**

(b) Given the periodic signal

$$x(t) = 2 + \frac{1}{2}\cos(t + 45°) + 2\cos(3t) - \sin(4t + 30°), \quad \textbf{(10)}$$

(i) Find the exponential Fourier series of x(t)

(ii) Sketch the magnitude and phase spectra as a function of ω.

(c) The triangular waveform x(t) of Fig. for Q. No. 1 (c) with period T = 4 and peak amplitude A = 4 is applied to a series combination of a resistor R = 100 Ω and an inductor L = 0.1 H. Determine the power dissipated in the resistor. **(15)**

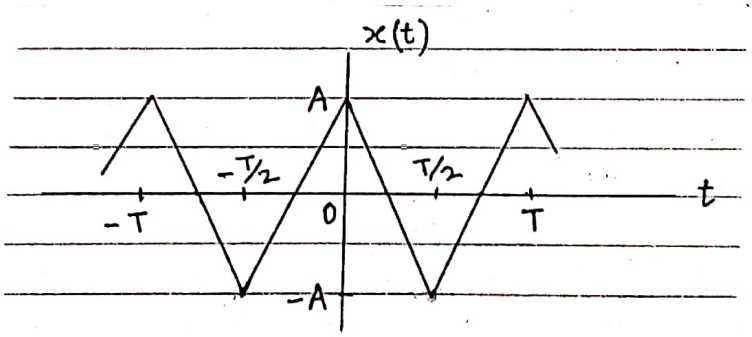


Fig. for Q. No. 1 (c)

2. (a) Prove that the energy of a signal x (t) is given by **(10)**

$$E = \frac{1}{2\pi}\int_{-\infty}^{\infty} |x(\omega)|^2 \, d\omega,$$

where X(ω) is the Fourier transform of x(t). Also, compute the energy of a sinc signal, sinc $(\frac{t}{2})$.

(b) The signal x(t) = exp [-αt] u(t) is input into a system with impulse response h(t) = sin (2t)/(πt). **(10)**

Contd P/2

EEE 211

Contd... Q. No. 2(b)

(i) Find the Fourier transform Y(ω) of the output.

(ii) For what value of α does the energy in the output signal equal one-half the input signal energy?

(c) A single-sideband, amplitude modulated signal is generated using the system shown in Fig. for Q. No. 2 (c). (i) Sketch the spectrum of y(t) for $\omega_f = \omega_m$. (ii) Find the mathematical expression for $h_f(t)$. Is it a realizable filter? **(15)**

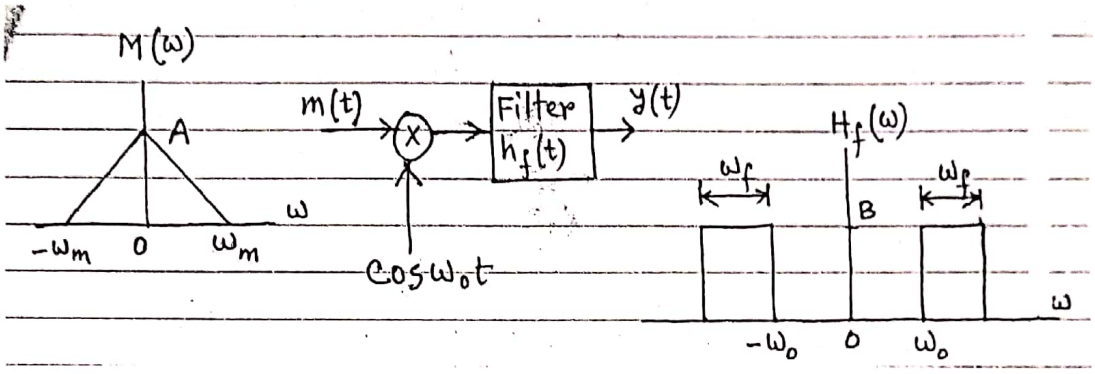


Fig. for Q. No. 2 (c)

3. (a) Show that the Fourier transform of a signal consisting of a train of impulses in time domain yields a train of impulses in frequency domain. Sketch both the signal and its spectrum. **(10)**

(b) Determine the Fourier transform of the signal x(t) = sin 1000 πt + cos 1200 πt. Sketch the spectrum X(ω). What should be the sampling period ~~that period~~ that would ensure no aliasing? **(10)**

(c) A periodic signal x (t) with period T = 2 has the Fourier coefficients

$$C_n = \begin{cases} 0, & n = 0 \\ 0, & n \text{ is even} \\ 1, & n \text{ is odd} \end{cases}$$

The signal is applied to a linear time-invariant continuous-time system with the magnitude and phase spectra shown in Fig. for Q. No. 3 (c). Determine h(t) and the system output. **(15)**

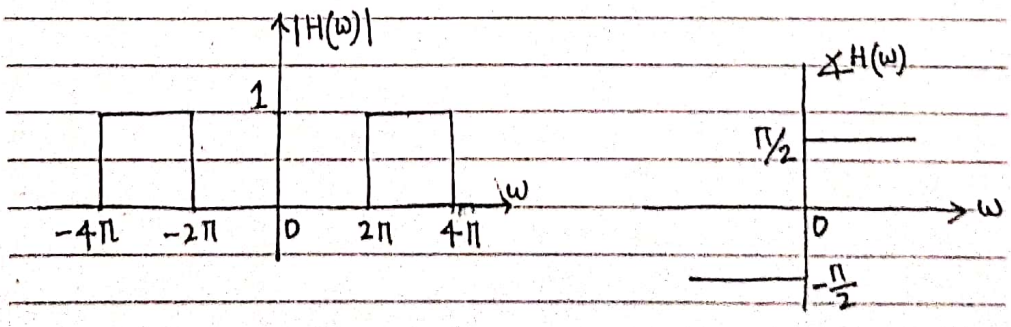


Fig. for Q. No. 3 (c)

Contd P/3

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4. (a) The input and the output of a causal LTI system are related by the differential equation $\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 2x(t)$. By using Fourier transformation,

(i) find the impulse response of the system,

(ii) find the output y(t) of the system if the input $x(t) = te^{-2t}u(t)$ **(17)**

b) For the mechanical system of Fig. for Q. No. 4(b), write the equation of motion in terms of the given mechanical quantities. Obtain the equivalent electrical equations and draw the f–i (force-current) and f-v (force–voltage) analogous electrical circuits. **(18)**

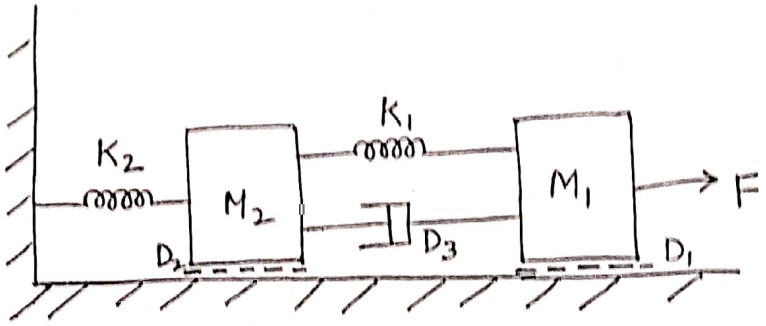


Fig. for Q. No. 4 (b)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) The following information is provided about a real signal x(t) with Laplace transform X(s). **(15)**

(i) X(s) has exactly two poles.

(ii) X(s) has no zeros in the finite s-plane

(iii) X(s) has a pole at $s = -1 + j$

(iv) $e^{2t}x(t)$ is not absolutely integrable.

(v) $X(0) = 5$

Determine X(s) and specify its region of convergence.

(b) Consider the circuit shown in Fig.for Q. No. 5 (b), where **(20)**

$$I_s(t) = \begin{cases} 2A, t \le 0 \\ 4e^{-2t}A, t \ge 0^+ \end{cases}$$

Determine $i_L(t)$ for $t \ge 0^+$ using Laplace transform method.

Contd P/4

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Contd... Q. No. 5(b)

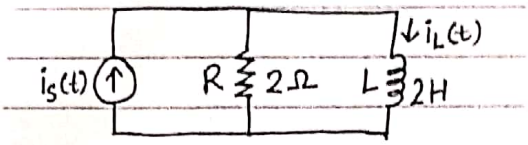


Fig. for Q. No. 5 (b)

6. (a) Determine whether the following signals are power or energy signals or neither. Justify the answer.

(i) $x_1(t) = \exp\left[j\left(2\pi + \pi/4\right)\right] + \exp[-2|t|]$,

(ii) $x_2(t) = [u(\cos t) - \delta(t - k\pi)]^{0.5}$, where $k = \pm 1, \pm 3, \pm 5, \ldots\ldots \infty$. **(15)**

(b) Consider the voltage waveform v(t) shown in Fig. for Q. No. 6 (b). **(20)**

(i) Express v(t) in terms of unit step and unit ramp functions.

(ii) Compute the derivative of v(t) and sketch its waveform.

(iii) Evaluate and sketch the even and odd components of the derivative of v(t)

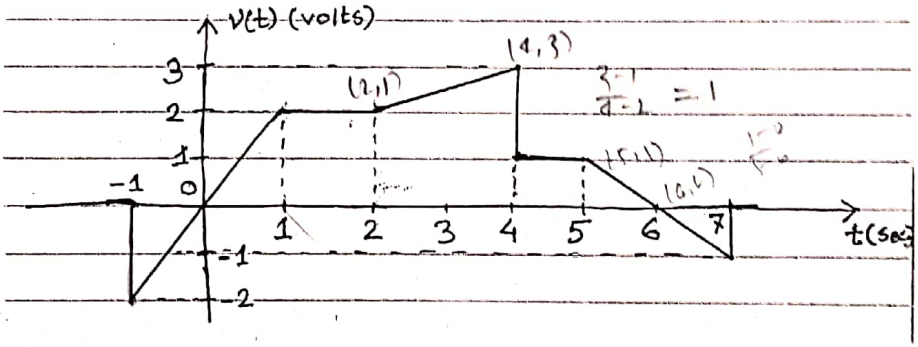


Fig. for Q. No. 6 (b)

7. (a) A linear system h(t) has the input-output as shown in Fig. for Q. No. 7(a). Determine whether the system is causal, time-invariant, and memory less. Also, find the output of the system for the input $x_1(t) = u(t) + u(t-1) - 2u(t-2)$ **(20)**

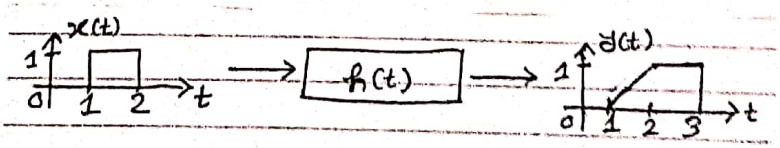


Fig. for Q. No. 7 (a)

Contd P/5

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Contd... Q. No. 7

(b) Consider an LTI system which has the impulse response of $h(t) = u(t) - u(t-1)$. If $x(t) = t^2\,[u(t) - u(t-3)]$ is applied at the input of the system, obtain the output $y(t)$ of the system using the graphical interpretation of convolution. Also, write the analytical expression of $y(t)$ **(15)**

8. (a) Consider an LTI system with input $x(t)$ and output $y(t)$ governed by the differential equation $\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + ky(t) = \frac{d^2x(t)}{dt^2}$, for $t \geq 0^+$

(i) Assume K is a real-valued constant and all the initial conditions to be zero. For what values of K is the system bounded-input bounded-output stable?

(ii) Find the zero-input response for the initial conditions, $y(0^+) = 0$ and $y'(0^+) = 1$. Assume $K = 2$ for this case. (15)

(b) Consider the RLC circuit shown in Fig. for Q. No. 8 (b). This circuit is excited with a voltage source of $x(t)$. Assuming the current in the circuit as the output and the current through the inductor, $v_1(t)$, and the voltage across the capacitor, $v_2(t)$, as the state variables, find the state space model of the circuit. Also, find a closed-form expression for the state transition matrix of the system. (20)

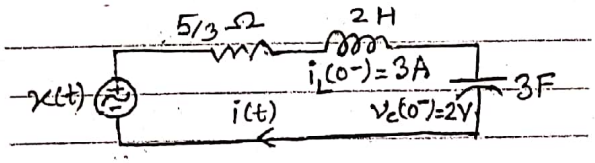


Fig. for Q. No. 8 (b)

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2017-2018

Sub : **EEE 211** (Continuous Signals and Linear Systems)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

Symbols have their usual meanings.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) What is meant by orthogonal Signals? For the functions shown in Fig. for Q. No. 1(a), (i) show that the functions form an orthogonal set, (ii) determine T that makes the three functions orthonormal. **(12)**

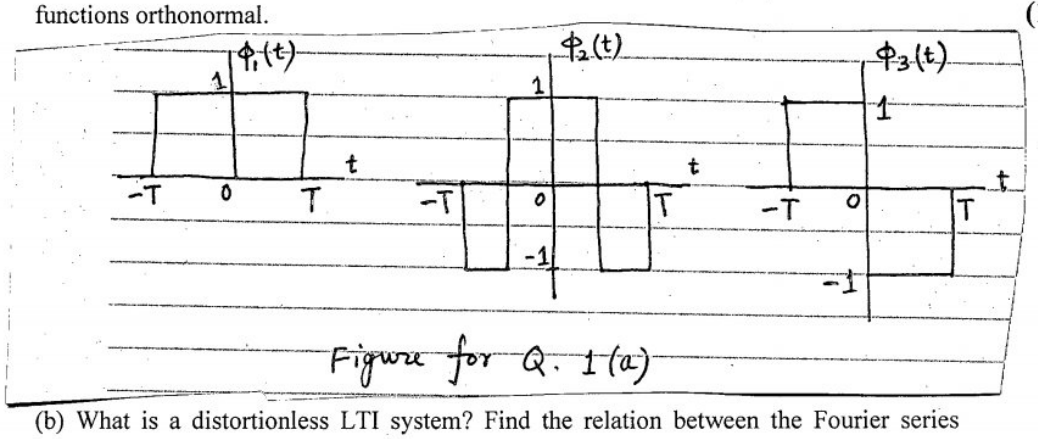


Figure for Q. 1(a)

(b) What is a distortionless LTI system? Find the relation between the Fourier series coefficients of input and output of a distortionless LTI system. **(8)**

(c) Suppose that we have an LTI system with input x(t), output y(t) and frequency response H(jw), where

$$H(j\omega)=\begin{cases}1 \ for \ |\omega| \le 3\pi \\ 0 \quad otherwise\end{cases}$$

Further suppose that the input is given by

$$x(t)=1+2\cos 2\pi t+\cos 4\pi t+\frac{1}{2}\cos 6\pi t\,.$$

Find the exponential Fourier series representation of x(t). Use this representation is order to find the response y(t) of the system to the input x(t). Plot the frequency spectra of x(t) and y(t). **(15)**

2. (a) Derive the perseval's relation of energy for an aperiodic signal. **(7)**

(b) For an electrical network system shown in Fig. for Q. No. 2(b),

$h(t)=\frac{1}{RC}e^{-1/RC}u(t)$. Using frequency domain method, find y(t) when x(t) = Vu(t). Also, plot h(t), x(t), and y(t). **(13)**

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Contd … Q. No. 2(b)

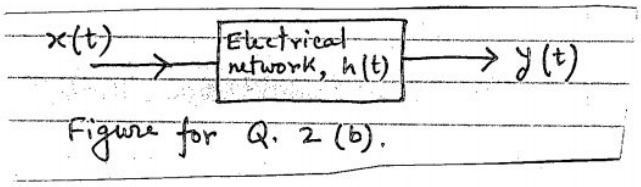


Figure for Q. 2 (b).

(c) In natural sampling, the signal x(t) is multiplied by a train of rectangular pulses, as shown in figure for Q. No. 2(c). (i) Find and sketch the spectrum of $x_s(t)$, (ii) can x(t) be recovered without any distortion? **(15)**

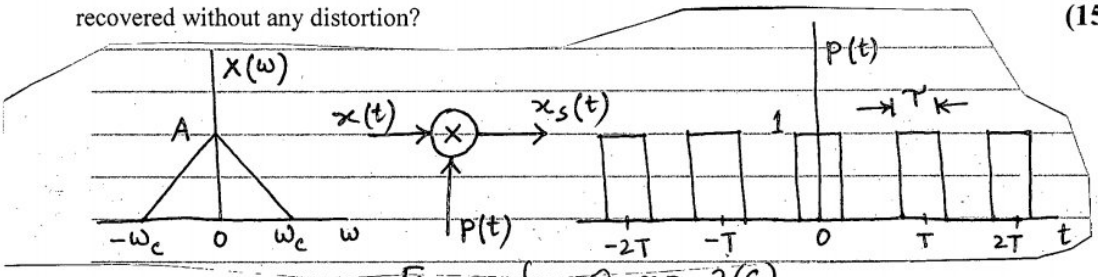


Figure for Q. no. 2(c)

3. (a) Find the s-domain equivalent circuit for the circuit shown in Fig. for Q. No. 3(a) and find y(t) when x(t) = (cost)u(t). Assume that $v_C(0^-) = 1$ V and $i_L(0^-) = 2$ A. **(17)**

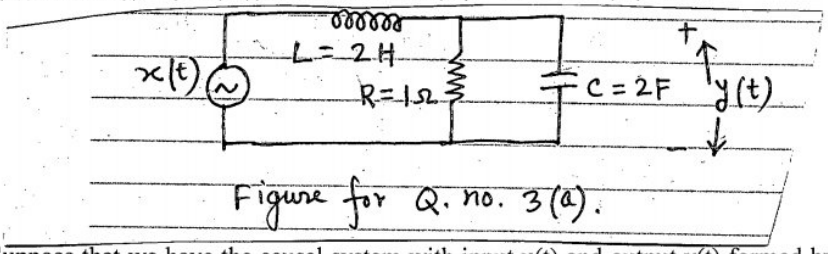


Figure for Q. no. 3(a).

(b) Suppose that we have the causal system with input x(t) and output y(t) formed by the interconnection of two causal LTI systems as shown in Fig. for Q. No. 3(b). The system functions be

$$H_1(s) = \frac{1}{s^2 + bs + (b-2)} \text{ and } H_2(s) = -1,$$

where b be a real constant. (i) Find the system function H(s) of the overall system, (ii) Determine the values of the parameter b for the which the system is stable. **(18)**

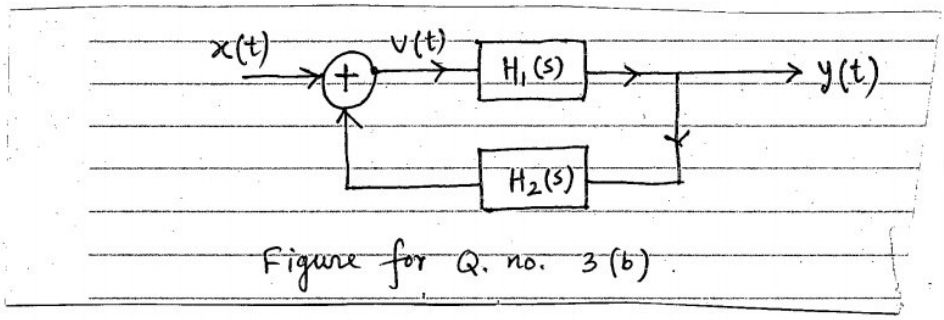


Figure for Q. no. 3 (b)

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EEE 211

4. (a) Calculate the trigonometric Fourier series coefficient for the signal

$$x(t) = 3 + cos\left(4t + \frac{\pi}{4}\right) + sin\left(10t + \frac{\pi}{3}\right) \quad \textbf{(12)}$$

(b) The model of an LTIC system is given by

$$y''(t) + 3y'(t) + 2y(t) = 2x'(t) + x(t)$$

Calculate the transfer function and the impulse response h(t) for the system. **(13)**

(c) Determine the Laplace transform of the periodic wave shown in Fig. for Q. No. 4(c).

If the period is 2, what is the ROC of the transform? **(10)**

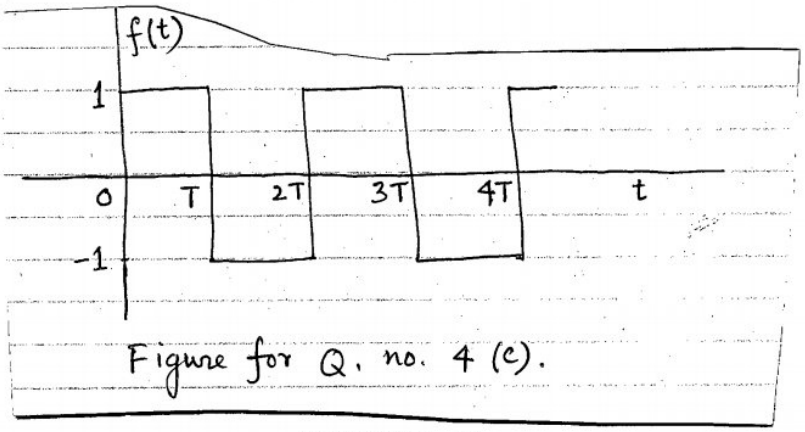


Figure for Q. no. 4 (c).

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE** questions.

5. (a) Consider the following signals

$$x_1(t) = 2\,cos\left(0.6\sqrt{\pi}t + \pi/6\right)$$
$$x_2(t) = sin\left(1.2\sqrt{\pi}t - \pi/3\right)$$

Is the signal x(t) = $x(t) = [x_1(t) + x_2(t)]^2$ periodic? In the case of periodic signal, determine the fundamental period. **(15)**

(b) Determine whether the following signals are power or energy signals or neither. Justify your answer. **(20)**

$$x_1(t) = x(4 - 1/2)\left[\delta(t + 3/2) - \delta(t - 3/2)\right]$$
$$x_2(t) = \sum_{k=-1}^{1} u(2x(t) - k)$$

Where the signal x(t) is given in Fig. for Q. 5(b).

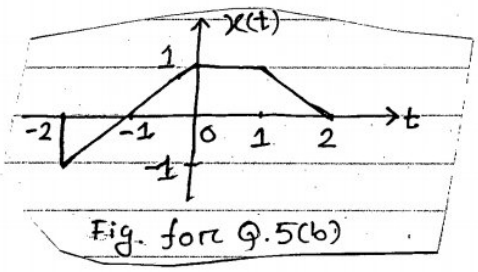


Fig. for Q.5(b)

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6. (a) Consider an LTI system whose response to the signal $x_1(t) = u(t) - u(t-1)$ is the signal $y_1(t)$ illustrated in Fig. for Q. 6(a). Determine and sketch the response of the system to the input $x_2(t)$ shown in Fig. for Q. 6(a). **(20)**

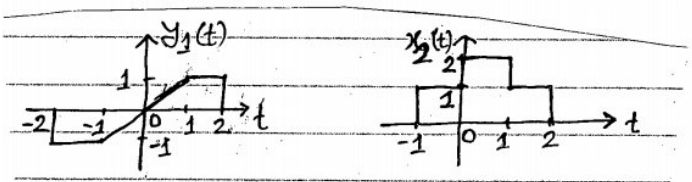


Fig. for Q. 6(a)

(b) Consider the system shown in Fig. for Q. 6(b). Determine whether it is (a) time-invarient, (b) memoryless, (c) causal, (d) invertible, (e) stable or not. **(15)**

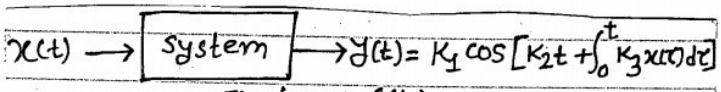


Fig. for Q. 6(b)

where, k_1, k_2, and k_3 are constants.

7. (a) The input and impulse response of an LTI system are denoted by x(t) and h(t), respectively, and are shown in Fig. for Q. 7(a). Using the graphical interpretation of convolution, obtain the output of the system, y(t). Also, write the analytical expression of y(t). **(20)**

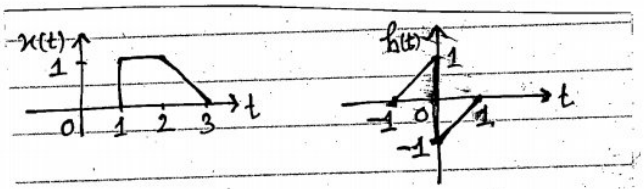


Fig. for Q. 7(a)

(b) The input signal of the LTI system shown in Fig. for Q. 7(b) is the following:

$x(t) = u(t) - u(t-2) + \delta(t+1)$

The impulse responses of the subsystems are $h_1(t) = te^{-t}u(t)$, $h_2(t) = e^{-2t}u(t)$. **(15)**

(i) Determine the impulse response h(t) of the overall system.

(ii) Find an equivalent system (same impulse response) configured as a parallel interconnection of two LTI subsystems.

(iii) Determine the output signal y(t).

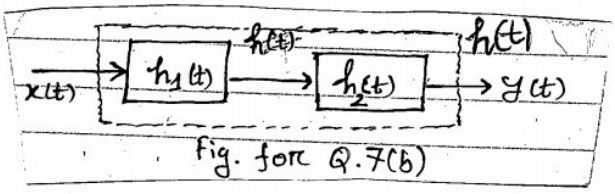


Fig. for Q. 7(b)

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8. (a) Consider the system simulation diagram of Fig. for Q. 8(a). **(20)**

(i) Find the differential equation of the system.

(ii) Using state-variable techniques, find the impulse response (h(t) for the system described by the differential equation obtained in (i).

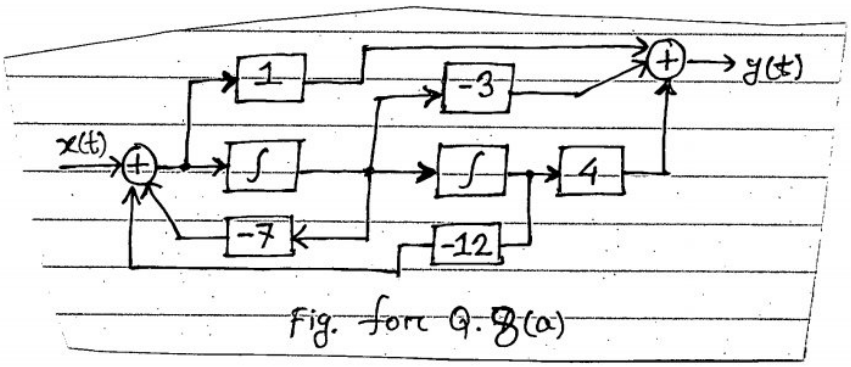


Fig. for Q. 8(a)

(b) For the circuit shown in Fig. for Q. 8(b), write the electrical and analogous mechanical differential equations by using force-current (f – i) analogy. Also draw the f – i analogous mechanical system for the circuit. **(15)**

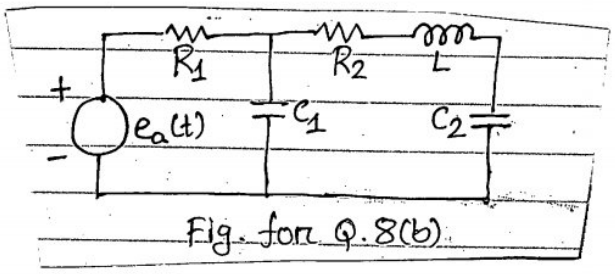


Fig. for Q. 8(b)

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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-2/T-1 B. Sc. Engineering Examinations 2016-2017

Sub : **EEE 211** (Continuous Signals and Linear Systems)

Full Marks : 280 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) Determine if the systems with the following impulse response are memoryless, causal, and stable: **(15)**

   (i) $h(t) = \delta(t) - \delta(t-2)$,

   (ii) $h(t) = 2\,\text{rect}(t/2)$,

   (iii) $h(t) = 2e^{-4t}u(t)$,

   (iv) $h(t) = [1 - e^{-4t}]u(t)$.

   (b) Determine the impulse response of the system shown in the figure for Q. No. 1(b), if $h_1(t) = e^{-2t}u(t)$, $h_2(t) = e^{-2t}u(t)$, $h_3(t) = e^{-t}u(t)$, $h_4(t) = \delta(t)$ and $h_5(t) = e^{-3t}u(t)$. Find the output $y(t)$ of the system if the input $x(t) = u(t)$. Also, comment or the stability of the system. **(20)**

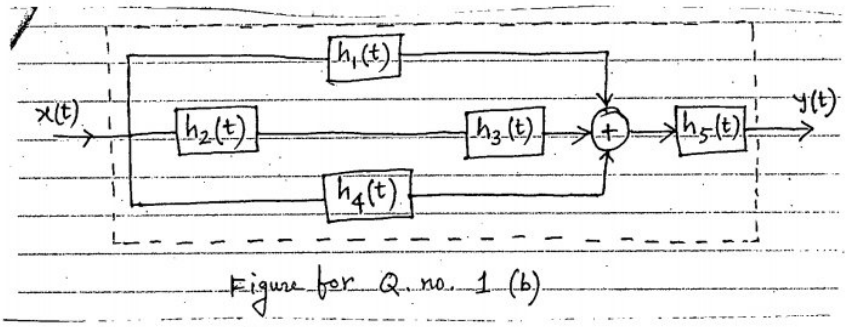


Figure for Q. no. 1 (b)

2. (a) State and explain the Dirichlet conditions of convergence for Fourier Series. **(10)**

   (b) The input to a system in given by x(t) as shown in the following figure for Q. No. 2(b). **(25)**

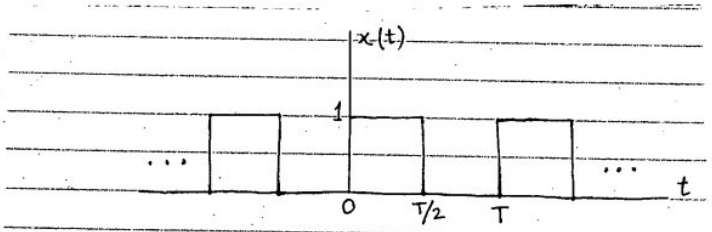


Figure for Q. no. 2 (b)

   (i) Find the exponential Fourier series of x(t).

   (ii) Calculate the average power in x(t) using parseval's theorem.

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Contd ... Q. No. 2(b)

(iii) What will be the Fourier series of $\frac{dx(t)}{dt}$?

(iv) If the frequency response of the system is given by $H(\omega)=\frac{1}{1+j\omega}$, find the Fourier series of the output y(t) of the system.

(v) Plot the magnitude and phase spectrum of x(t).

3. (a) For the functions shown in figure for Q. No. 3(a), (i) show that the functions form an orthogonal set, (ii) Determine T that makes the three functions orthonormal, and (iii) Express the signal x(t) in terms of functions of the orthogonal set: **(15)**

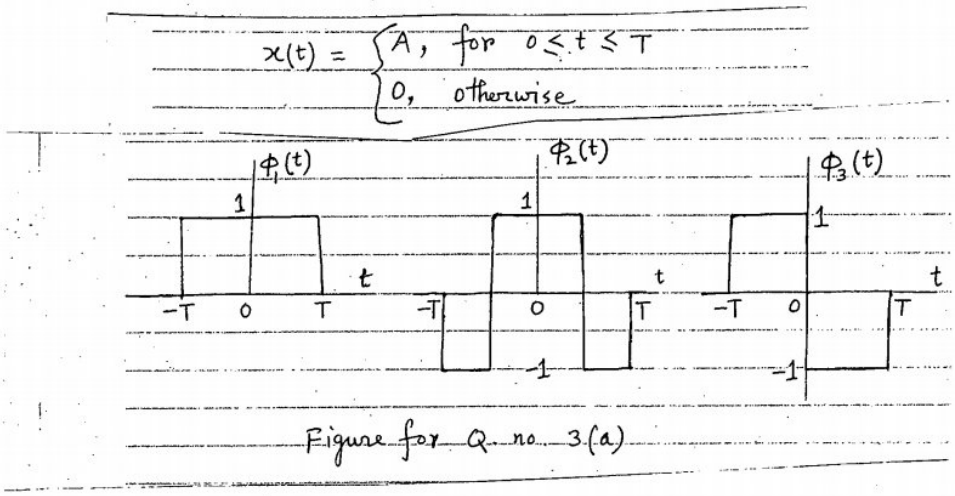


Figure for Q. no. 3(a)

(b) What is a distortionless LTI system? Find the relation between the Fourier Series coefficients of input and output of a distortionsless LTI system. **(10)**

(c) Find the impulse response of the initially relaxed system shown in figure for Q. No. 3(c): **(10)**

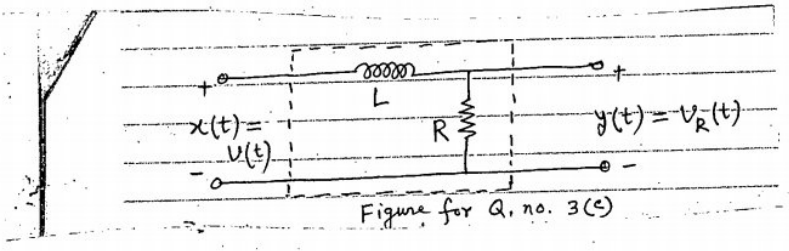


Figure for Q. no. 3(c)

Also, find the step response of the system.

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4. (a) Find a state space representation of the circuit shown in figure for Q. No. 4(a), assuming that the outputs are the currents flowing in R_1 and R_2. **(15)**

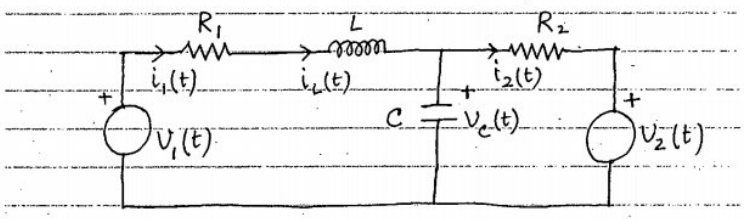


Figure for Q. no. 4(a)

(b) Write the properties of state transition matrix. Find an expression of the complete solution of response y(t) from state equations for continuous time LTI system and show that the impulse response is given by **(10)**

$$h(t)=\begin{cases}\overline{c\phi}(t)\overline{b}+d\delta(t), t\geq 0\\ 0, \text{ otherwise}\end{cases}$$

where the variables have usual meanings.

(c) Represent the system with transfer function: **(10)**

$$H(s)=\frac{s^2-3s+2}{s^3+6s^2+11s+6}$$

by simulation diagram using canonical forms.

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE** questions.

5. (a) Sketch and label the signal

$$x(t) = (t + 1)\,[u(t + 2) - u(t)].$$ **(10)**

(b) The signal y(t) as shown in Fig. for Q. 5(b) is obtained by the transformation

$$y(t) = x(-2t + 6).$$ **(15)**

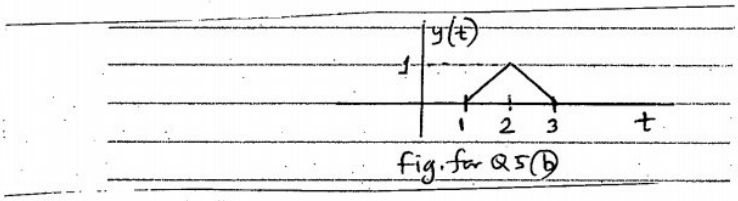


fig. for Q5(b)

Sketch the signal x(t).

(c) Determine whether the signal $x(t) = \cos(100\,\pi t + \pi/3) + \sin(40\pi\, t + 2\pi/3)$ is periodic or not. If it is periodic, find the fundamental period of the signal. **(10)**

Contd P/4

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EEE 211

6. (a) Determine the Fourier transform of the rectangular pulse x(t)=rect(t/τ). Use this result to find the Fourier transform of the triangular pulse defined as **(12)**

$$y(t)=\Delta(t/\tau)=\begin{cases}1-|t|/\tau, & |t|\le\tau\\ 0, & |t|>\tau\end{cases}$$

using the convolution property of Fourier transform.

(b) Determine the energy of the signal **(11)**

$$x(t)=\frac{\sin(2\pi\times10(t-1/40))}{\pi(t-1/40)}\cos(2\pi\times15t).$$

(c) Consider the system depicted in Fig. for Q. No. 6(c) where **(12)**

$$\rho(t)=\sum_{n=-\infty}^{\infty}\delta\left(t-\frac{n\pi}{\omega_B}\right)$$

$$h(t)=\frac{\sin(3\omega_B/2)}{\pi t}$$

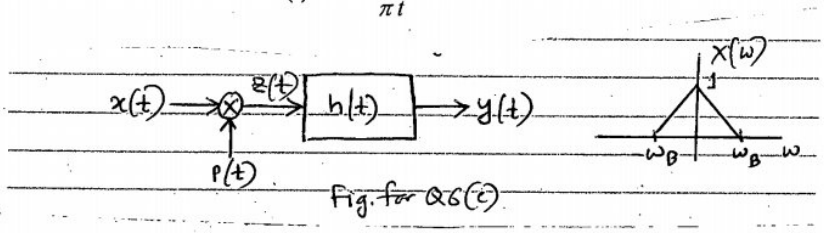


Fig. for Q6(c)

The input to the system has the spectrum shown in the same figure. Sketch the frequency spectrum of p(t), z(t), h(t) and y(t).

7. (a) Determine the bilateral and unilateral Laplace transform of the signal **(11)**

$$x(t)=e^{at}u(-t),\ a>0.$$

(b) The Laplace transform of the signal x(t) that is zero for t < 0 is **(12)**

$$X(s)=\frac{s^3+2s^2+3s+2}{s^4+2s^3+2s^2+2s+2}$$

Determine the Laplace transform of the signal

$$y(t)=tx(t)+\frac{dx(t)}{dt}+\int_0^t x(\tau)d\tau.$$

(c) Consider the system shown in Fig. for Q. 7(c). **(12)**

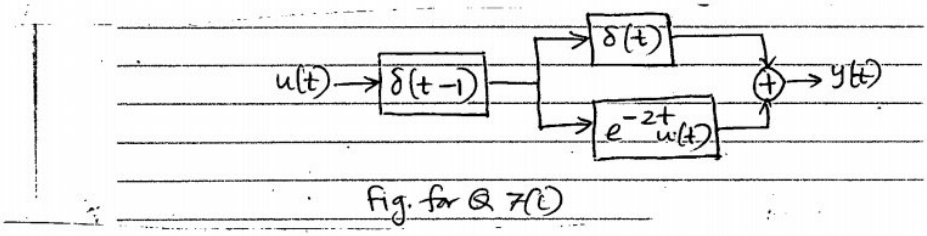


Fig. for Q 7(c)

Determine Y(s) and y(t).

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8. (a) Determine the impulse response h(t) of the ideal band stop filter whose frequency response is shown in Fig. for Q. No. 8(a). **(12)**

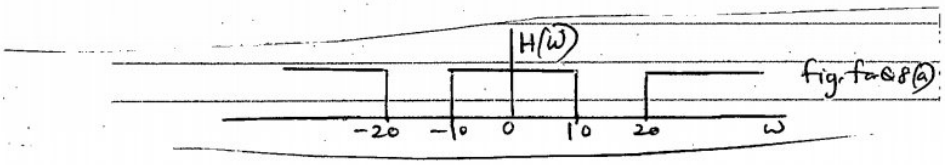


Fig. for Q 8(a)

(b) Consider the RLC circuit shown in Fig. For Q. No. 8(b). **(12)**

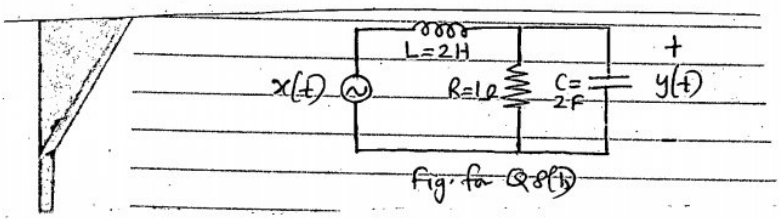


Fig. for Q 8(b)

Determine the transfer function H(s) = Y(s)/X(s) and impulse response h(t) of the RLC circuit.

(c) For the mechanical system shown in Fig for Q. No. 8(c), draw the f – i (force – current) analogous electrical circuit and write the equation of motion in terms of the given mechanical quantities. **(11)**

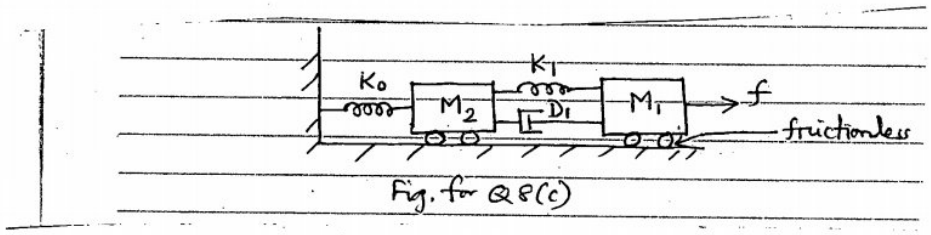


Fig. for Q 8(c)

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L-3/T-1/EEE **Date : 24/01/2017**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2015-2016

Sub : **EEE 301** (Continuous Signals and Linear Systems)

Full Marks: 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Symbols have their usual meanings.

1. (a) Determine if the following signals are periodic or aperiodic. If periodic, find the period: **(20)**

   (i) $x(t)=3\cos(4t+\pi/3)$,

   (ii) $x(t)=[\cos(2t-\pi/3)]^2$,

   (iii) $x(t)=\sum_{n=-7}^{7} e^{-(2t-n)}$,

   (iv) $x(t)=e^{j(\pi t-1)}$.

   (b) Find whether the following signals are power or energy signals. **(15)**

   (i) $x(t)=e^{-t}\cos(3t)u(t)$,

   (ii) $x(t)=u(t)-2u(t-2)+u(t-5)$,

   (iii) $x(t)=te^{-t}u(t)-\delta(t-2k);\quad k=0,\pm1,\pm2\cdots\pm\alpha$

2. (a) Evaluate the following: **(20)**

   (i) $\int_{-\alpha}^{\alpha} f(\tau)\delta(t-\tau)\,d\tau$,

   (ii) $\int_{-\alpha}^{\alpha}(t^2+3t+2)\delta(-1-t)\,dt$,

   (iii) $\int_{-\alpha}^{\alpha} e^{(t-1)}\sin\left\{\frac{\pi}{2}(t-5)\right\}\delta(t+3)\,dt$,

   (iv) $\left(\frac{jw+2}{w^2+9}\right)\delta(w)$,

   (v) $\frac{1}{jw+2}\delta(w-3)$

   (vi) $\int_{-5}^{5} u_2(1-\tau)\cos 2\pi\tau d\tau;\quad u_2(t)=\partial^2 u(t)/dt^2$.

   (b) For the following system, find if it is (i) linear, (ii) time-invariant, (iii) causal (iv) memoryless, and (v) stable: **(15)**

$$y(t)=\sin\{x(t)+x(t-2)\};\quad t\geq 0$$

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3. (a) A linear time-invariant (LTI) system is described by the following input-output relation: **(20)**

$$y''(t)+6y(t)+9y(t)=2x'(t)+8x(t)$$

(i) Draw the simulation diagram of the system for first canonical form (ii) Write the state-equation of the system, (iii) Find the state-transition matrix of the system, (iv) Determine the stability of the system.

(b) The unit impulse response of an LTI system is given by **(15)**

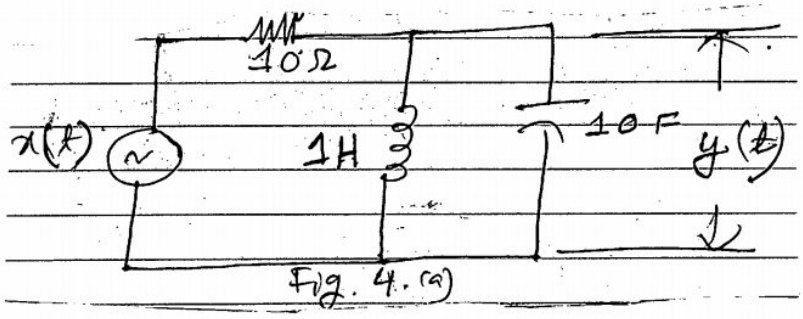
$$h(t)=\left[e^{-3t}-e^{-2t}\right]u(t)$$

Find the zero-state responses of the system for the inputs (i) $u(t)$ and (ii) $e^{-t}u(t)$.

4. (a) The Fourier series of a certain periodic signal is given by **(20)**

$$x(t)=3+\sqrt{3}\cos 2t+\sin 2t+\sin 3t-\frac{1}{2}\cos\left(5t+\pi/3\right)$$

Determine the output voltage $y(t)$ for the following circuit, where $x(t)$ is the input supply of the circuit. Assume zero-initial conditions.



Is the circuit distortionless?

(b) $x(t)$ and $y(t)$ are periodic signals with period T = 1/2, and given by $x(t)=\cos(4\pi t)$ and $y(t)=\sin(4\pi t)$. Find the Fourier series coefficients of $z(t)$ using the multiplication property, where $z(t)=x(t)\,y(t)$. **(15)**

SECTION-B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Signal $x(t)$ has the Fourier transform **(10)**

$$X(w)=\frac{2+j4w+w^2}{3+j4w-w^2}$$

Find the Fourier transform of

(i) $x(t)*x(t-1)$ (ii) $x(-2t+1)e^{-j2t}$

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EEE 301

Contd... Q. No. 5(a)

(b) Consider the system shown in Fig. for Q. No. 5(b). The systems $h_1(t)$ and $h_2(t)$, respectively, have frequency responses: **(15)**

$$H_1(w)=\frac{1}{2}[H_0(w-w_0)+H_0(w+w_0)]$$

$$H_2(w)=-\frac{1}{2j}[H_0(w-w_0)-H_0(w+w_0)]$$

Draw the magnitude spectra of $x_1(t)$, $x_2(t)$, $y_1(t)$, $y_2(t)$ and $y(t)$.

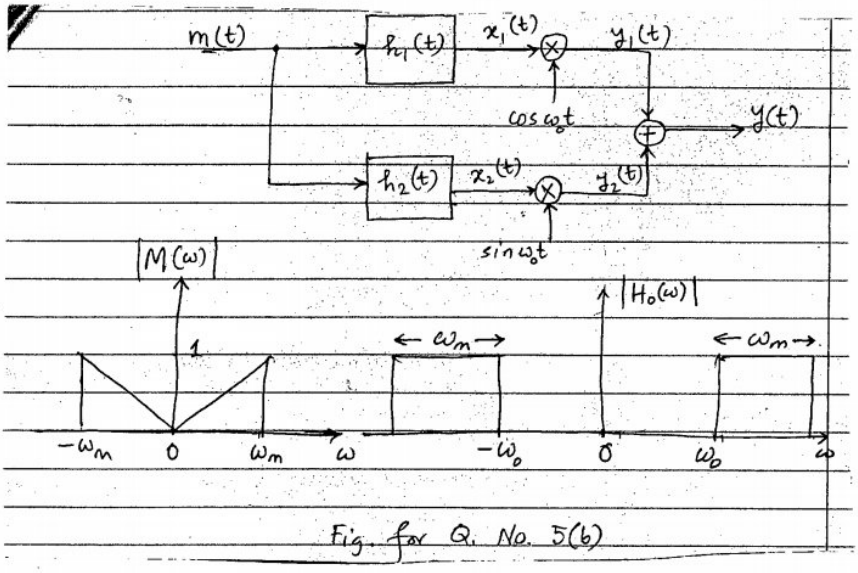


Fig. for Q. No. 5(b)

(c) Sketch the magnitude spectrum $|X(w)|$ for $x(t)=\frac{sin(t)\,sin(t/2)}{\pi t^2}$. **(10)**

6. (a) For the signal $x(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT)$ with period T, find Fourier series representation and Fourier transform representation. Also find the power-density spectrum of $x(t)$. **(10)**

(b) Let $x(t)$ and $y(t)$ be low-pass signals with bandwidths of 150 Hz and 350 Hz, respectively, and let $z(t)=x(t)\,y(t)$. The signal $z(t)$ is sampled using an ideal sampler at intervals of T_s secs. **(12)**

(i) What is the maximum value that T_s can take without introducing aliasing?

(ii) If $x(t)=sin\left(\frac{300\pi t}{\pi t}\right)$, $y(t)=sin\left(\frac{700\pi t}{\pi t}\right)$

Sketch the magnitude spectrum of the sampled signal for.

(i) T_s = 0.5 ms and (ii) T_s = 2 ms

(c) The input and the output of a causal LTI system are related by the differential equation **(13)**

$$\frac{d^2y(t)}{dt^2}+6\frac{dy}{dt}+8y(t)=2x(t)$$

(i) Find frequency response H(w).

(ii) Using frequency domain approach find the response of this system if $x(t)=te^{-2t}u(t)$.

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7. (a) For an LTI system with input $x(t)$ and output $y(t)$, the state space parameters are **A**, **b**, **c**, and d. The state space representation is as follows **(17)**

$$\mathbf{v}'(t) = \mathbf{A}\,\mathbf{v}(t) + \mathbf{b}\,x(t)$$
$$y(t) = \mathbf{c}\,\mathbf{v}(t) + d\,x(t)$$

Derive an expression of system transfer function H(s). Also find H(s) and h(t) for the following parameters.

$$\mathbf{A} = \begin{bmatrix} -2 & 3 \\ -1 & 2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -1 & -1 \end{bmatrix}, \quad d = 1.$$

(b) Using Laplace transform method, find y(t) for the following differential equation **(13)**

$$y''(t) + 5y'(t) + 6y(t) = 2e^{-t}u(t), \quad y'(0^-) = -2 \text{ and } y(0^-) = 3$$

(c) Explain briefly the D'Alembert's principle. **(5)**

8. (a) For the circuit shown in Fig. for Q. No. 8(a), find y(t) considering $v_c(0^-) = 1$ volt, $i_L(0^-) = 2$ amperes, and $x(t) = u(t)$. Use Laplace transform method and incorporate the initial energy for the inductor and the capacitor in your transformed model. **(15)**

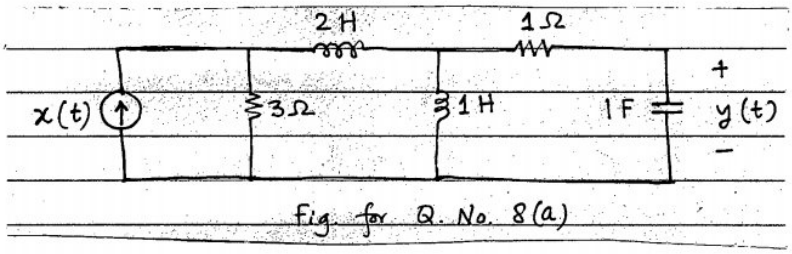


Fig. for Q. No. 8(a)

(b) For the Fig. for Q. No. 8(b), find Y(s) and final value of y(t). **(10)**

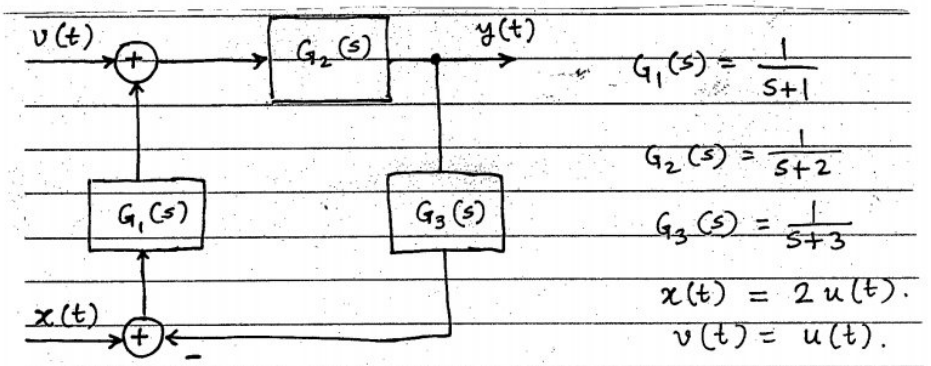


Fig. for Q. No. 8(b)

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Contd... Q. No. 8

(c) For the mechanical system shown in Fig. for Q. No. 8(c), draw the f-v analogous electrical circuit. Also write the equations of motion in terms of the given mechanical quantities **(10)**

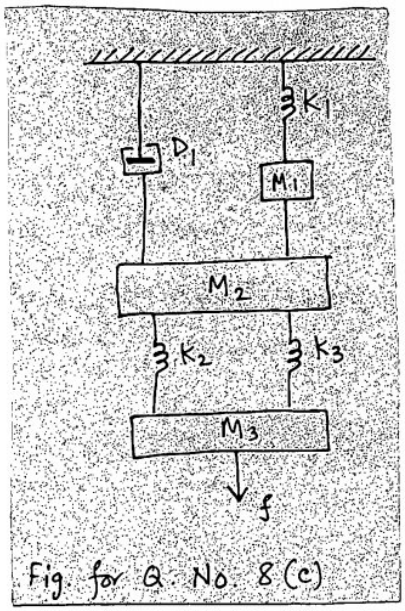


Fig. for Q. No. 8 (c)

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BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2014-2015

Sub : **EEE 301** (Continuous Signals and Linear Systems)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Symbols bear their usual meanings.

1. (a) Consider the following signal **(9)**

$$x(t) = u(\cos t) - \delta(t - k\pi) \qquad k = 0, \pm 1, \pm 2, \ldots\ldots, \pm \infty$$

Is the signal periodic? Give reason. If periodic, determine the fundamental period. Finally, sketch the signal.

(b) Determine whether the following signal is power or energy signal or neither. Justify our answer. **(6)**

$$x(t) = \sum_{k=-4}^{k=4} \delta(3t - k) \sin\left(\frac{\pi t}{3}\right) \mathrm{sgn}(t)$$

(c) Consider a signal given by **(20)**

$$x(t) = \begin{cases} -t+1 & -1 \le t < 0 \\ t & 0 \le t < 2 \\ 2 & 2 \le t < 3 \\ 0 & \text{othwrwise} \end{cases}$$

Find the analytical expressions of the odd and even parts of the signal. What are the average values of these parts of the signal?

2. (a) Determine whether the following systems are **(18)**

- linear/nonlinear
- time-invariant/time variant
- causal/non-causal
- memoryless/with memory
- invertible/non-invertible
- stable/unstable

(i) $y(t) = \sin(2t) \log_{10} [x(t)]$

(ii) $y(t) = e^{-j\omega t} \int_{-\infty}^{\infty} x(\tau) e^{-j\omega\tau} \, d\tau$

(b) Determine the impulse response of the system that has an input-output relation given by **(17)**

$$y'(t) + 10y(t) = 2x(t)$$

Find and sketch the output of the system, when the input is

$$x(t) = (-1)^k \delta(t - 5k), \qquad k = 0, \pm 1, \pm 2$$

Assume that $\lim_{t \to 5} h(t) = 0$, where h(t) is the impulse response of the system.

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3. An LTI system is defined by the following differential equation **(35)**

$$y''(t) + 6y'(t) + 8y(t) = x'(t) - x(t)$$

(i) Draw the simulation diagram of the system using the first canonical form.

(ii) Find the state equation of the system.

(iii) Determine the state-transition matrix of the system.

(iv) Comment on the stability of the system.

(v) Find the transformation matrix to convert the first canonical form into the second.

4. (a) The Fourier series coefficients of a periodic signal with period 20 ms are **(18)**

$$C_n = \frac{1}{2}\left[1 - (-1)^n\right] \qquad n = 0, \pm 1, \pm 2, \ldots\ldots\ldots \pm \infty$$

(i) Does this represent a real signal? Justify your answer.

(ii) From the expression of C_n, deduce the time-domain signal. Sketch the signal.

(b) A voltage x(t) is applied to the circuit shown in Fig. for Q. 4(b). The Fourier series coefficients of x(t) are given by **(17)**

$$C_n = \frac{1}{n^2 + 1} \exp\left[\frac{jn\pi}{3}\right] \qquad n = 0, \pm 1, \pm 2, \ldots, \pm \infty$$

Determine the amplitude of the first four harmonics of the output signal y(t). What is the average value of the output signal?

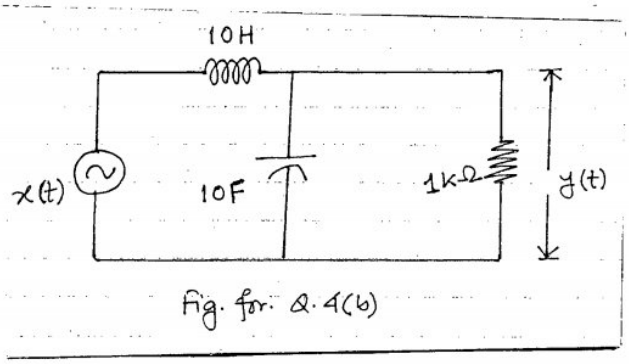


Fig. for Q. 4(b)

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) A single-sideband amplitude-modulated signal is generated using the system shown below **(15)**

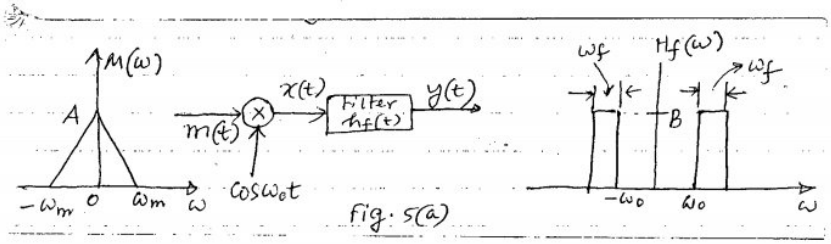


fig. 5(a)

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EEE 301

Contd ... Q. No. 5(a)

(i) Sketch the spectrum of y(t) for $\omega_f = \omega_m$

(ii) Write the mathematical expression of $h_f(t)$

(b) Determine the Fourier transform of the pulse $x(t) = \alpha\ \text{rect}(\alpha t/\tau)$, $\alpha > 0$. Comment on this Fourier transform as $\alpha \to \infty$ and as $\alpha \to 0$. **(7)**

(c) The input signal $x(t) = \exp[-\alpha t]\ u(t)$ is applied to a system whose impulse response h(t) is $\sin(2t/(\pi t))$. Find the Fourier transform of the output. **(13)**

6. (a) The input to the system shown in Fig. 6(a) has the spectrum shown below: **(16)**

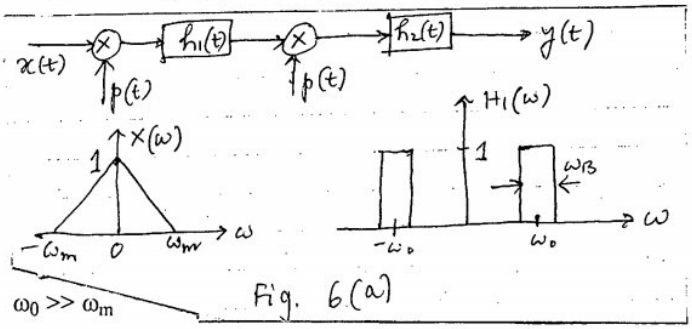


Fig. 6(a)

Let $p(t) = \cos\omega_0(t)$, $\omega_0 >> \omega_m$

Find the spectrum $Y(\omega)$ of the output if $h_2(t) = \dfrac{\sin \omega_B t}{\pi t}$, Assume that $\omega_m > \omega_B$ and $\omega_m \leq \omega_B$.

(b) In natural sampling, the signal x(t) is multiplied by a train of rectangular pulse as shown in Fig. 6(b). **(15)**

(i) Find and sketch the spectrum of $x_s(t)$.

(ii) Can x(t) be recovered without any distortion? Explain.

(c) State D' Alembert's principle. **(4)**

7. (a) Using Laplace transform method, find y(t) for the following differential equation **(15)**

$y''(t) + 4y'(t) + 3y(t) = \exp[-3t]\ u(t)$

$y(0^-) = 0,\ y'(0^-) = 1$

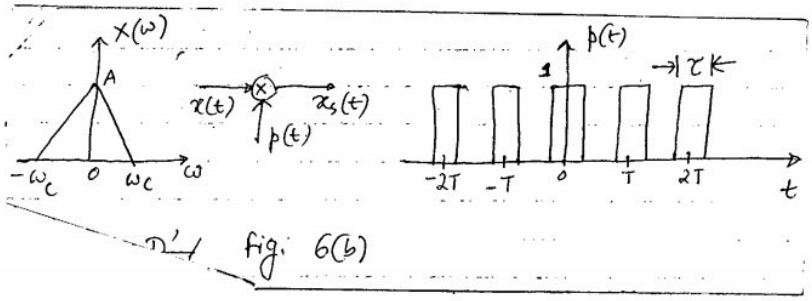


Fig. 6(b)

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EEE 301

Contd ... Q. No. 7

(b) Find the bilateral Laplace transform and ROC of **(7)**

$x(t) = \text{expt}[bt]\, u(-t)$

(c) For the system shown in Fig. 7(c), find Y(s). **(13)**

8. (a) For the circuit shown in ig. 8(a), $v_c(0^-) = 1$ volt, $i_L(0^-) = 2$ amperes and $x(t) = u(t)$. **(13)**
Draw the Laplace transformed model incorporating the initial energy for the inductor and the capacitor. Find the expression of Y(s) and y(t).

(b) Consider the system described by **(18)**

$$v'(t) = \begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix} v(t) + \begin{bmatrix} 1 \\ 3 \end{bmatrix} x(t)$$

$$y(t) = [-1 \quad -1]\, v(t) + 2x(t)$$

State variable $v(t) = \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$ and $v(0^-) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

(i) Find H(s) and comment on the stability of the system.

(ii) Find the step response of the system.

(c) Draw the analogous electrical circuit considering suitable analogy for the system shown in Fig. 8(c). **(4)**

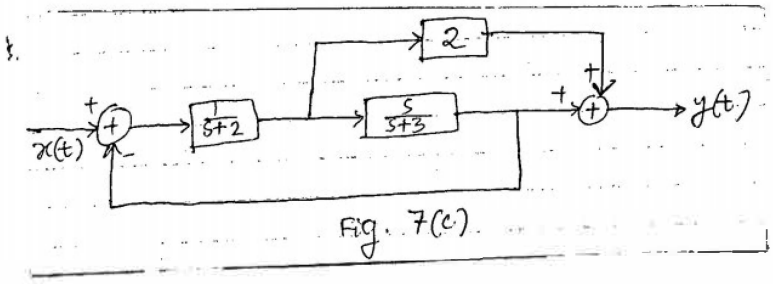


Fig. 7(c)

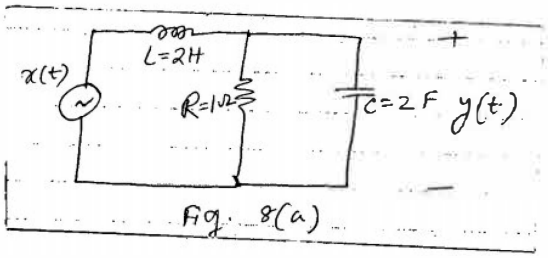


Fig. 8(a)

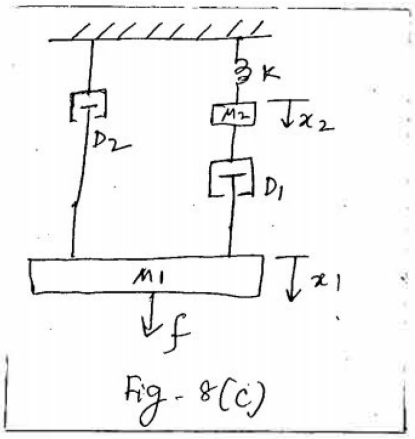


Fig - 8(c)

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L-3/T-1/EEE Date : 05/07/2015

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2013-2014

Sub : **EEE 301** (Continuous Signals and Linear Systems)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Symbols have their usual meanings

1. (a) Determine whether $x(t) = x_1(t) + x_2(t)$ is periodic when **(18)**

(i) $x_1(t) = 4\cos(\pi t)$, $x_2(t) = -\sin(3\pi t + \pi/2)$

(ii) $x_1(t) = 4\cos(2t)$, $x_2(t) = 4\cos(2\pi t)$

If periodic, find the corresponding periods.

(b) Find if the following signals represent a delta function: **(17)**

(i) $x(t) = \underset{\epsilon \to 0}{\text{Lt}} \frac{1}{\pi} \cdot \frac{\epsilon}{t^2 + \epsilon^2}$

(ii) $p(t) = \underset{\epsilon \to 0}{\text{Lt}} \left(\frac{1}{\pi}\right)\left(\frac{\sin \epsilon t}{t}\right)$

2. (a) Determine whether the following signals are power or energy signals: **(18)**

(i) $x_1(t) = u(t) + 5u(t-1) - 2u(t-2)$

(ii) $x_2(t) = 2\sin\left(\frac{3\pi}{8}\right)t \cos\left(\frac{2\pi}{3}\right)t$

(b) An analogy system has the following input-output relation: **(17)**

$$y(t) = \int_0^t e^{-(t-\tau)} x(\tau)\, d\tau; \qquad t \geq 0$$

$$= 0; \qquad \text{otherwise}$$

Here, x(t) and y(t) represent the input and output of the system, respectively.

Find if the system is linear and time-invariant (LTI). Next determine its impulse response without any computation. Also, determine if the system is BIBO stable.

3. (a) Find if the periodic signals $x_1(t) = \tan(2\pi t)$ and $x_2(t) = \sin(2\pi/t)$ have a convergent Fourier series representation. **(18)**

(b) Find the exponential Fourier series representation of the following signal: **(17)**

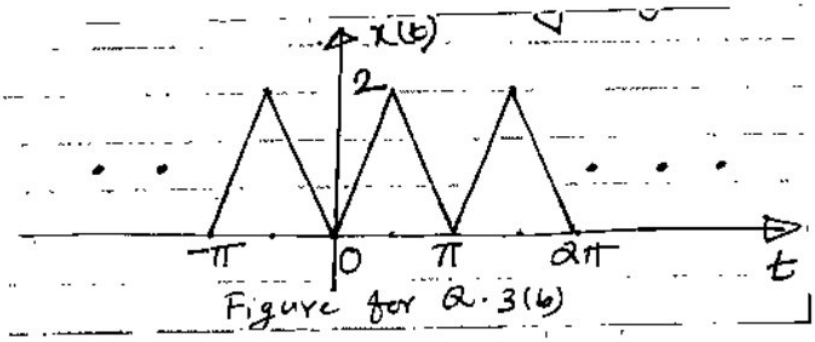


Figure for Q. 3(b)

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4. (a) For the following circuit, write down the state equations. Consider x(t) and y(t) as the input and output for the equations, respectively. Find if the circuit is stable using the state matrix A. **(18)**

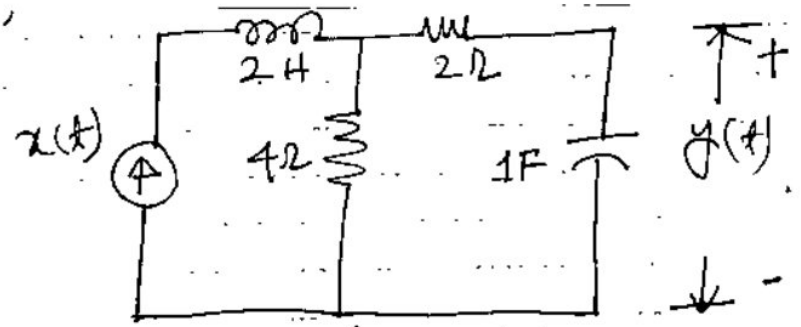


Figure for Q.4(a)

(b) The periodic signal $x(t) = B + A\cos(\omega_0 t + \theta)$ is applied as input of the following circuit, where $\omega_0 = 2\pi/T_0$, T_0 being the period, and θ represents the phase of the input signal. **(17)**

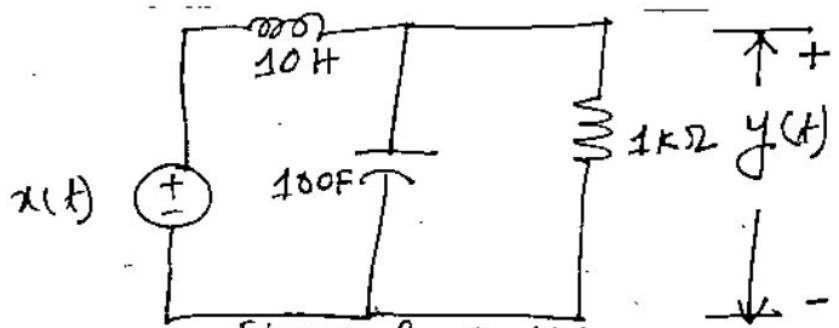


Figure for Q.4(b)

Find the output voltage y(t).

SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) For the circuit shown in Fig. for Q. No. 5(a). **(18)**

$v_c(0^-) = \frac{1}{2}$ volt, $i_L(0^-) = -1$ ampere and $x(t) = u(t)$

Draw the transformed model incorporating the initial energy for the inductor and the capacitor. Find the expressions of Y(s) and y(t).

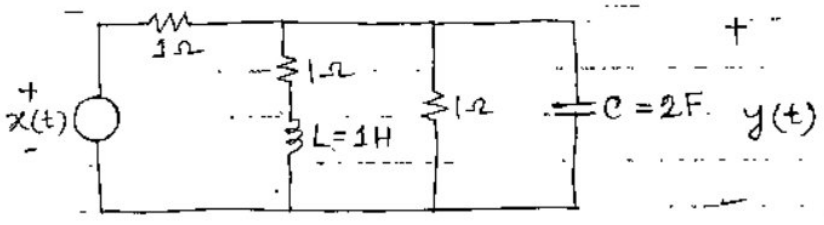


Fig. for Q. no. 5(a)

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Contd ... Q. No. 5

(b) $\dot{v}(t) = \begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} v(t) + \frac{1}{2}\begin{bmatrix} 1 \\ 2 \end{bmatrix} x(t)$ **(17)**

$y(t) = [1 \quad 1]\, v(t) + x(t)$

State variable $v(t) = \begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix}$ and $v(0^-) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(i) Find H(s) and comment on stability of the system.

(ii) Find step response of the system.

6. (a) Using Laplace transform method find y(t) for the following differential equation **(13)**

$y''(t) + 4\, y'(t) + 3\, y(t) = u(t)$

$y(0^-) = 2, \quad y'(0^-) = 1$

(b) Find the bilateral Laplace transform and the ROC of **(7)**

$x(t) = \cos(at)\, u(-t)$

(c) **(15)**

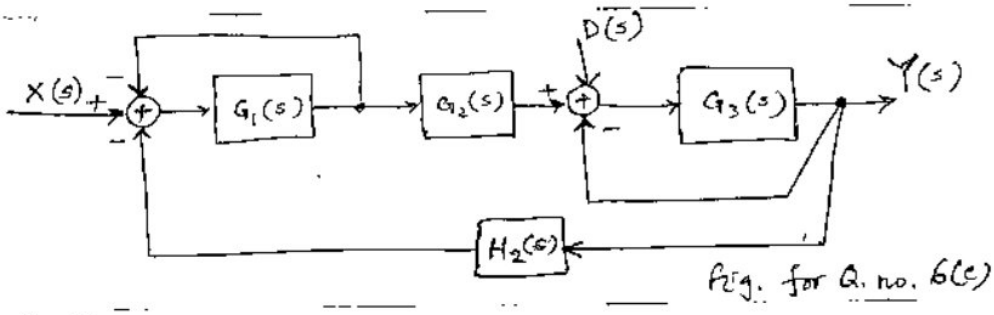


Fig. for Q. no. 6(c)

Consider,

$X(s) = \frac{1}{s}, \quad D(s) = \frac{1}{s}, \qquad$ find Y(s) and $\lim_{t \to \infty} y(t)$

$G_1(s) = \frac{1}{s+1}, \quad G_2(s) = \frac{1}{s+2}, \quad G_3(s) = \frac{1}{s+3}, \quad H_2(s) = s + 4.$

7. (a) Consider the amplitude modulation system shown below **(15)**

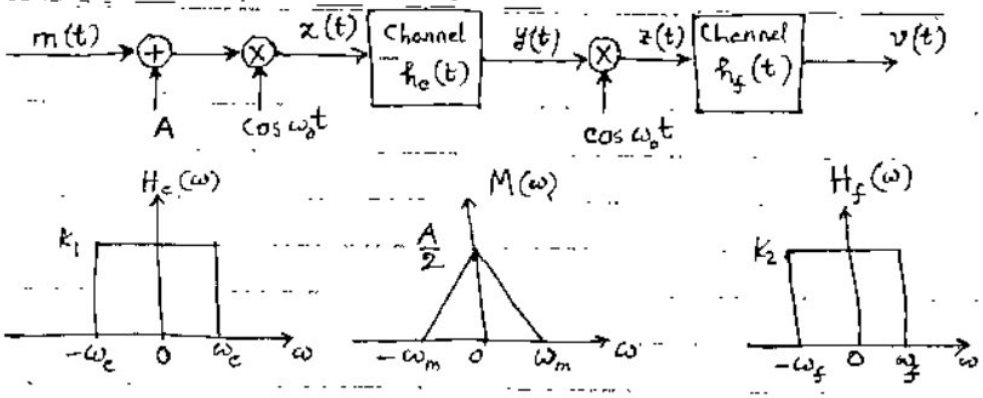


Fig. for Q. no. 7(a)

Sketch X(ω), Y(ω), Z(ω) and V(ω) for

$\omega_0 - \omega_m \le \omega_c < \omega_0 ; \qquad \omega_f = 2\omega_0 - \omega_m$

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Contd ... Q. No. 7

(b) Show that $F\left\{\text{sinc}\left(\frac{\omega_B t}{2\pi}\right)\right\} = \frac{2\pi}{\omega_B}\,\text{rect}\left(\frac{\omega}{\omega_B}\right)$. **(7)**

(c) The signal $x(t) = \left[\frac{1}{12}e^{-t/4} + \frac{1}{6}e^{-t/2}\right]u(t)$ is input into a system with impulse response $h(t) = e^{-t/3}\,u(t)$. Find the Fourier transform $Y(\omega)$ of the output and thus determine $y(t)$. **(13)**

8. (a) The signal $x(t)$ is multiplied by a periodic impulse train $p(t)$ with period $\frac{1}{8}$ m sec and the resulting signal $x_s(t)$ is filtered to obtain $y(t)$. **(17)**

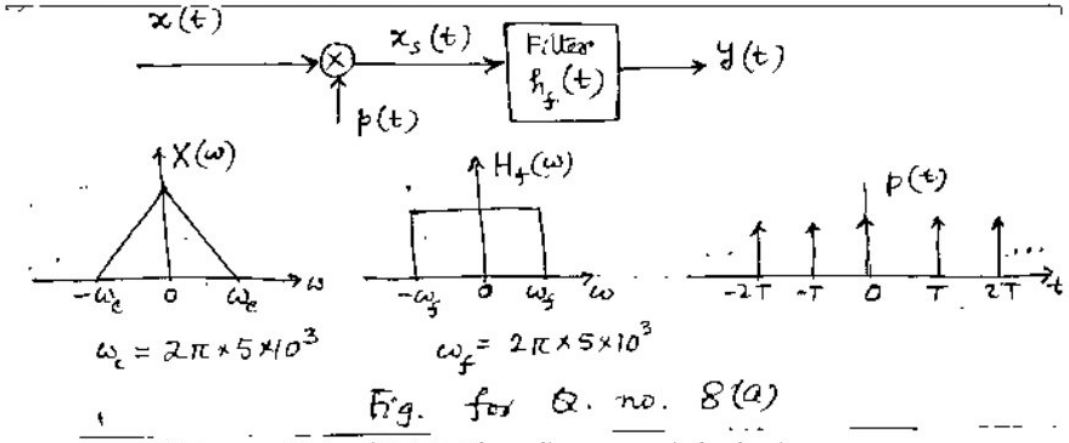


Fig. for Q. no. 8(a)

(i) Determine $P(\omega)$ and $X_s(\omega)$. (Show all necessary derivations)

(ii) Sketch $X_s(\omega)$ and $Y(\omega)$.

(iii) Can $x(t)$ be recovered from $x_s(t)$. If yes, specify necessary modification of the filter shown in the figure.

(b) **(10)**

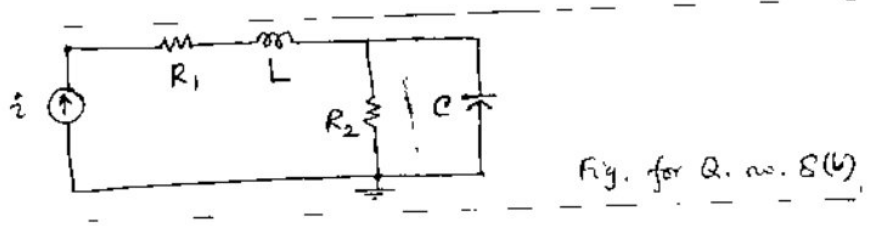


Fig. for Q. no. 8(b)

Using force-current analogy draw the analogous mechanical circuit. Assume that the system was initially relaxed. Write down necessary equations to justify your answer.

(c) Draw the analogous electrical circuit considering suitable analogy. **(8)**

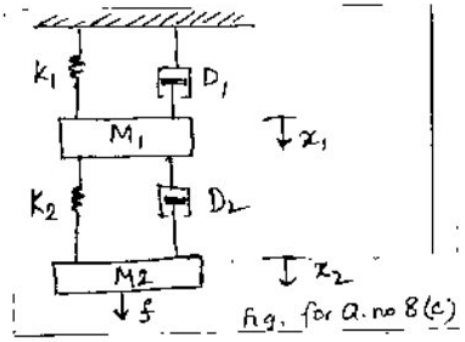


Fig. for Q. no 8(c)

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A. Nazeem 7/6/14

L-3/T-1/EEE Date: 07/06/2014

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2012-2013

Sub: **EEE 301** (Continuous Signals and Linear Systems)

Full Marks: 210 Time: 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this section. Answer any **THREE**.

Symbols bear their usual meanings.

1. (a) Consider the following signal **(12)**

$$x(t) = \alpha\, \delta(\sin t) + (1-\alpha)\, \delta(t - k\pi)$$

$$0 < \alpha < 1 \qquad k = \pm 1, \pm 3, \pm 5, \ldots\ldots \infty$$

Is the signal periodic? Give reason. In the case of periodic signal, determine the fundamental period. Finally, sketch the signal, when $\alpha = 0.5$.

(b) Determine whether the following signal is power or energy signal or neither. Justify the answer. **(12)**

$$x(t) = \sum_{k=-4}^{k=+4} \delta(3t - k) \sin^2\left(\frac{\pi t}{3}\right)$$

(c) Let $x_o(t)$ and $x_e(t)$ represent the odd and even parts of an input signal x(t), respectively. The output of a system is such that **(11)**

$$y(t) = x_e(t) - x_o(t)$$

Is the system invertible? If yes, specify the inverse.

2. (a) Determine whether the following systems are **(18)**

- linear or nonlinear
- time invariant or time variant
- causal or non causal
- memory less of with memory
- invertible or noninvertible
- stable or unstable

(i) $y(t) = \int_{-\infty}^{2t-3} x(\tau)\, d\tau$

(ii) $y(t) = \frac{1}{2}|x(t)| - \pi t$

(b) Determine the output y(t) of the system shown in Fig. for Q. No. 2(b), if the input **(17)**

$$x(t) = (-1)^k\, \delta(t - 10k), \qquad k = 0, \pm 1, \pm 2, \ldots\ldots \infty$$

The impulse responses are given as

$h_1(t) = \delta(t + 1)$ $h_2(t) = \delta(1 - 1)$

$h_3(t) = \delta(t)$ $h_4(t) = \text{rect}(t|2)$

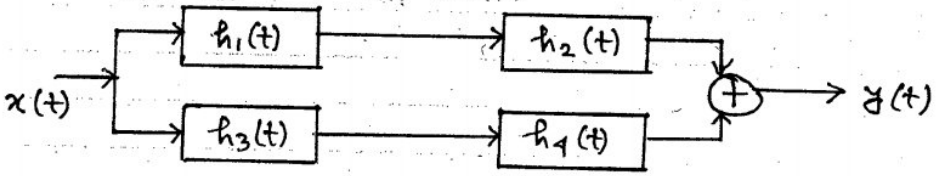


Fig. for Q. No. 2(b)

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3. An LTI system is defined by the following differential equation **(35)**

$$y'''(t) + 3y''(t) - y'(t) - 2y(t) = 3x''(t) - x(t)$$

(i) Draw the simulation diagram of the system using the canonical form-II.

(ii) Why integrators are used in the simulation diagram instead of differentiators?

(iii) Find the state equation of the system.

(iv) Determine the state transition matrix of the system using the Cayley-Hamilton method.

(v) Describe the significance of state transition matrix in view of evaluating the impulse response of the system and determining the stability of the system.

4. (a) Three orthogonal basis functions ϕ_1, ϕ_2 and ϕ_3 in an interval $0 \le t \le 3$ are given below: **(17)**

$$\phi_1(t) = \begin{cases} 1 & 0 \le t \le 2 \\ 0 & 2 < t \le 3 \end{cases} \qquad \phi_2(t) = \begin{cases} 1 & 0 \le t \le 1 \\ -1 & 1 < t \le 2 \\ 0 & 2 < t \le 3 \end{cases}$$

$$\phi_3(t) = \begin{cases} 0 & 0 \le t \le 2 \\ 1 & 2 < t \le 3 \end{cases}$$

Express x(t) shown in Fig. for Q. No. 4(a) in terms of $\phi_1(t)$, $\phi_2(t)$ and $\phi_3(t)$.

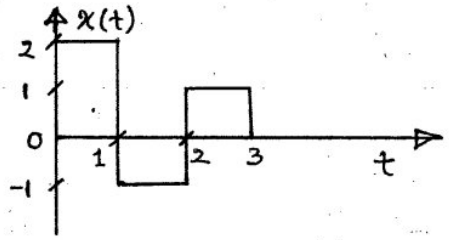
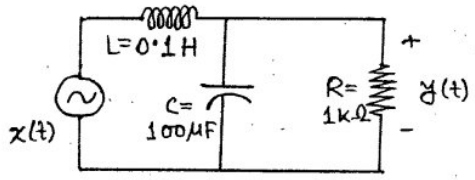


Fig. for Q. No. 4(a)

(b) Consider the circuit shown in Fig. for Q. No. 4(b). The input x(t) of the circuit is the half-wave rectified signal shown in this figure. Find the amplitude of the second and fourth harmonics of the output y(t). **(18)**



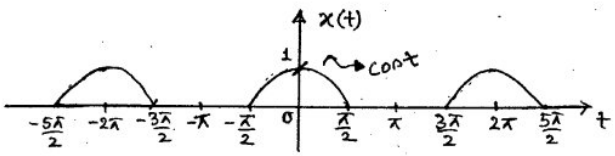


Fig. for Q. No. 4(b)

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SECTION – B

There are **FOUR** questions in this section. Answer any **THREE**.

5. (a) Find the Fourier transforms of the following signals – (i) $\delta(t)$, (ii) 1, (iii) $e^{j\omega_0 t}$, (iv) $\cos \omega_0 t$, (v) $\sin \omega_0 t$. **(12)**

(b) Consider the three related signals shown in Fig. 5(b). Determine $X(\omega)$ and using Fourier transform properties, compute $Y_1(\omega)$ and $Y_2(\omega)$ from $X(\omega)$. **(15)**

(c) Show that convolution in time domain is equivalent to multiplication in the frequency domain. **(8)**

6. (a) Write down the frequency response of ideal low-pass filter with cutoff frequency ω_c. Compute the impulse response of the filter and draw it. Comment on causality of the filter. **(8)**

(b) Consider the system shown in Fig. 6(b), where **(15)**

$$x(t) = \frac{\sin(\omega_B t/2)}{\pi t},$$

$$p(t) = \sum_{n=-\infty}^{\infty} \delta\left(t - \frac{n\pi}{\omega_B}\right),$$

$$h(t) = \frac{\sin(3\omega_B t/2)}{\pi t}.$$

Find $X(\omega)$, $P(\omega)$, $Z(\omega)$, $H(\omega)$, $Y(\omega)$, $S(\omega)$ and draw each of those.

(c) Consider the system shown in Fig. 6(c). The systems $h_1(t)$ and $h_2(t)$, respectively, have frequency responses **(12)**

$$H_1(\omega) = \frac{1}{2}[H_0(\omega - \omega_0) + H_0(\omega + \omega_0)]$$

$$H_2(\omega) = \frac{-1}{2j}[H_0(\omega - \omega_0) - H_0(\omega + \omega_0)]$$

Draw $H_1(\omega)$, $H_2(\omega)$, $X_1(\omega)$, $X_2(\omega)$ and $Y(\omega)$.

7. (a) consider the signal x(t) given in Fig. 7(a). Determine the Laplace transform of $|x(t)|$. **(10)**

(b) State the final-value theorem and prove it. **(5)**

(c) Find the impulse response of the system whose input and output are $e^{-2t}\,u(t)$ and $(e^{-t} - 3e^{-2t})\,u(t)$, respectively. **(5)**

(d) Draw (i) f-v analogous electrical circuit and (ii) f-i analogous electrical circuit for the mechanical system shown in Fig. 7(d). If the applied velocity, u(t) = 5 m/sec along with $M = \frac{1}{6}$ kg, $D = \frac{1}{5}$ N-sec/m, K = 1 m/N, then determine the force f(t) on the mass M [assume that the system was initially relaxed]. **(15)**

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8. (a) (i) Find H(s) of a system described by **(20)**

$$v'(t) = A\,v(t) + b\,x(t)$$

$$y(t) = c\,v(t) + d\,x(t)$$

(ii) Evaluate H(s) for the case –

$$A = \begin{bmatrix} -3 & 4 \\ -2 & 3 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$c = \begin{bmatrix} -1 & -1 \end{bmatrix}, \qquad d = 2$$

with $v(\bar{0}) = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

Comment on stability of the system. Also find the step response of the system.

(b) For the circuit shown in Fig. 8(b), let $v_c(\bar{0})$ = 1 volt, $i_L(\bar{0})$ = 2 amperes and x(t) = u(t),

(i) draw the transformed model, (ii) find the expressions of Y(s) and y(t). **(15)**

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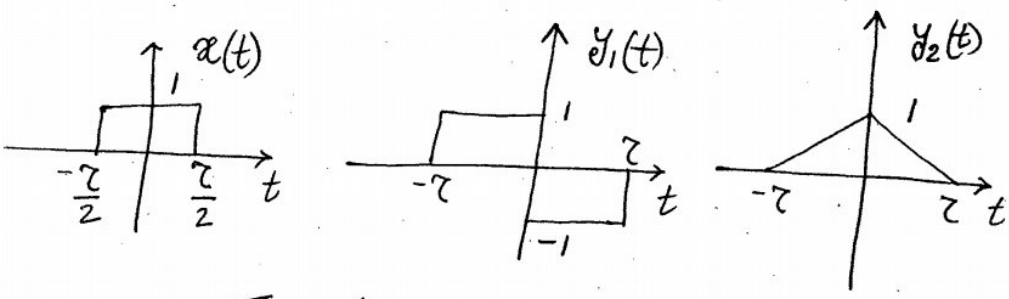


Fig. for Q. no. 5(b)

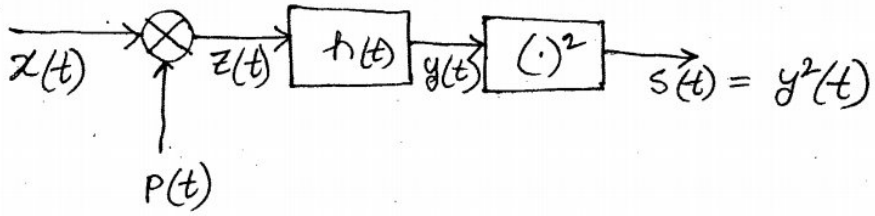


Fig. for Q. no. 6(b)

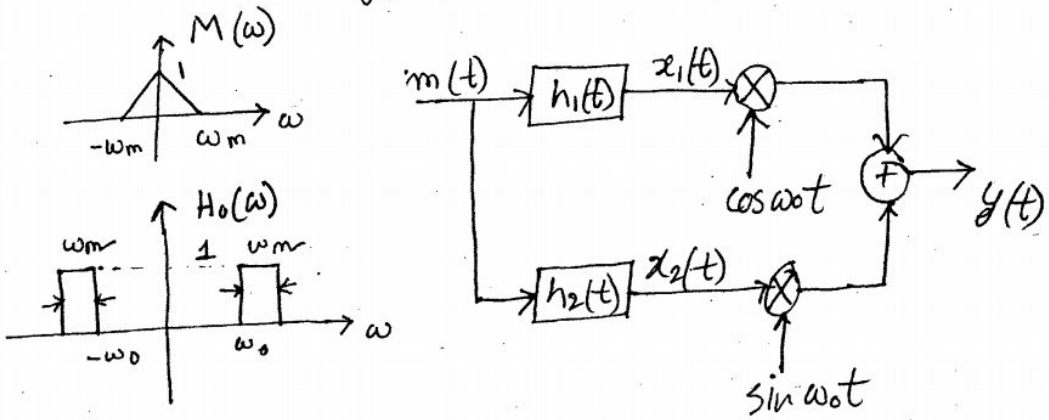


Fig. for Q. no. 6(c)

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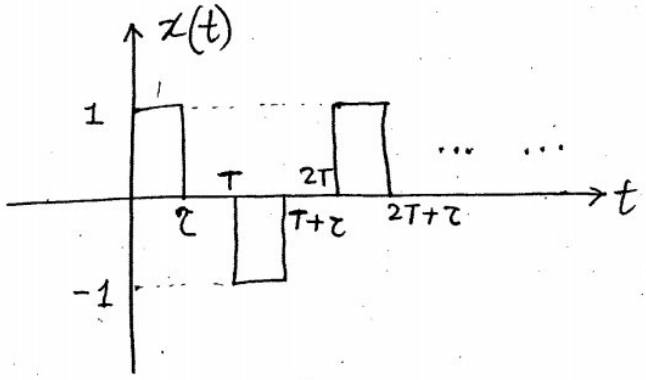


Fig. for Q. no. 7(a)

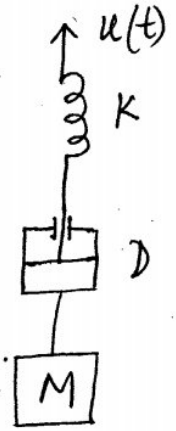


Fig. for Q. no. 7(d)

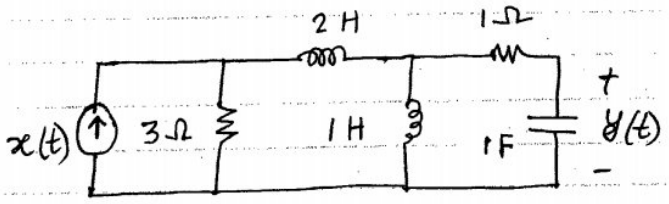


Fig. for Q. no. 8(b)

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L-3/T-1/EEE Date : 06/10/2013

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2011-2012

Sub : **EEE 301** (Continuous Signals and Linear Systems)

Full Marks : 210 Time : 3 Hours

The figures in the margin indicate full marks.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) The following information are given about a signal x(t): **(18)**

(i) x(t) is real and odd.

(ii) x(t) is periodic with period T = 2 and has Fourier co-efficients c_K.

(iii) $c_K = 0$ for $|K|>1$

(iv) $\frac{1}{2}\int_0^2 |x(t)|^2 dt = 1$

Specify two different signals that satisfy these conditions.

(b) Consider a continuous-time LTI system whose frequency response function is **(17)**

$$H(\omega) = \int_{-\alpha}^{\alpha} h(t)e^{-j\omega t}dt = \frac{\sin(4\omega)}{\omega}.$$

If the input to this system is a periodic signal

$$x(t) = 2 + x_1(t) * \sum_{k=-\alpha}^{\alpha} \delta(t-8k)$$

where $x_1(t) = \begin{cases} 1, & 0 \le t < 4 \\ -1, & 4 \le t < 8, \end{cases}$

Then using the Fourier Series method determine the corresponding system output y(t).

2. (a) Let

$$g(t) = x(t)\cos^2 t * \frac{\sin t}{\pi t},$$

where '*' denotes the convolution operator. Assuming that x(t) is real and X(ω) = 0 for $|\omega| \ge 1$, find the impulse response h(t) of an LTI system that can map x(t) to g(t), showing the relationship between G(ω) and X(ω). **(17)**

(b) In the system shown in Fig. for Q. 2(b) the sampling signal is an impulse train with alternating sign. The Fourier transform of the input signal is as indicated in the figure. For $T < \pi/(2\omega_m)$, determine a system that will recover x(t) from y(t), if there exists. Also plot the Fourier transform of $x_p(t)$ and y(t) for $T < \pi/(2\omega_m)$. **(18)**

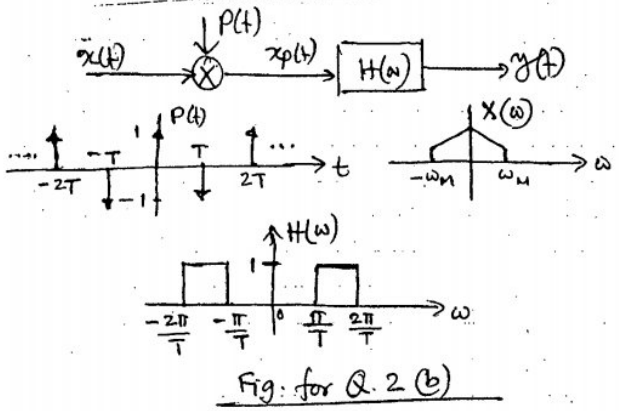


Fig. for Q. 2 (b)

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EEE 301

3. (a) Suppose that we are given the following information about an LTI system: **(18)**

(i) The system is causal

(ii) The system function is rational and has only two poles, at s = – 2 and s = 4.

(iii) If x(t) = 1, then y(t) = 0.

(iv)The value of the impulse response at $t = 0^+$ is 4.

Determine the system function H(s) of the system.

(b) The direct form II structure of a continuous time LTI system in the s-domain is shown in Fig. for Q. 3(b). Specify the differential equation that characterizes the system. Also compute the steady-state and transient responses resulting from the input x(t) = 2 cos 4t u(t), with no initial energy at t = 0. Use the unilateral Laplace transform method. **(17)**

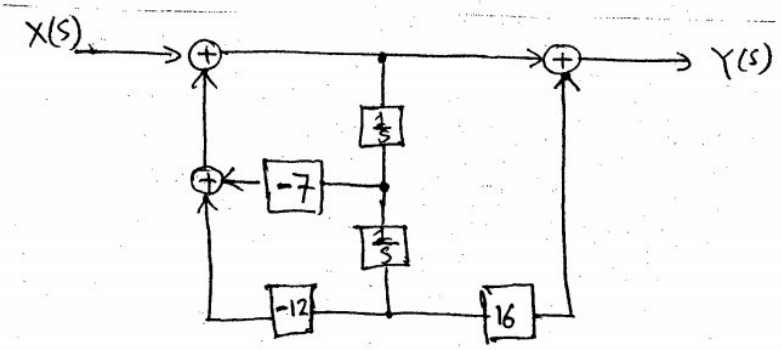


Fig. for Q. 3 (b)

4. (a) The input-output relationship of a non-causal continuous time LT1 system with one causal pole can be described by the differential equation **(18)**

$$\frac{d^2y(t)}{dt^2}+\frac{dy(t)}{dt}-6y(t)=\frac{dx(t)}{dt}-x(t)$$

If the input to the system, x(t) is $x(t) = e^{-t}u(t) + e^{t}u(-t-2)$,

determine the ROC of Y(s), the bilateral Laplace transform of y(t), and the system response y(t). Use the bilateral Laplace transform method.

(b) An ideal (– π/2) radian (or – 90°) phase shifter is defined by the frequency response

$$H(\omega)=\begin{cases} e^{-j\pi/2}, & \omega>0 \\ e^{j\pi/2}, & \omega<0 \end{cases}$$

Find the impulse response h(t) of this phase shifter and, the output y(t) where $x(t) = \cos\omega_0 t$ using the Fourier transform method. **(17)**

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) The input output state-space representation of an N-dimensional LTI system is given by **(7)**

$$V'(t) = A\,V(t) + \mathbf{b}x(t) \ldots (1)$$

$$y(t) = \mathbf{c}V(t) + d\,x(t),$$

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EEE 301

Contd ... Q. No. 5

where A, $\mathbf{b}$, $\mathbf{c}$ and d are state-space coefficients and V(t) is N-dimensional state vector. Show that the solution of state differential equation (1) is

$$V(t) = \Phi(t-t_0)V_0 + \int_{t_0}^{t} \Phi(t-\tau)\mathbf{b}x(\tau)d\tau$$

where state transition matrix $\Phi(t) = \exp[At]$ and initial state $V_o = V(t_o)$.

(b) For the RLC circuit shown in Fig. for Q. No. 5(b), considering voltage across the capacitor and current through the ~~indicator~~ inductor as two state variable $v_1(t)$ and $v_2(t)$, respectively find the input output state-space representation. **(10)**

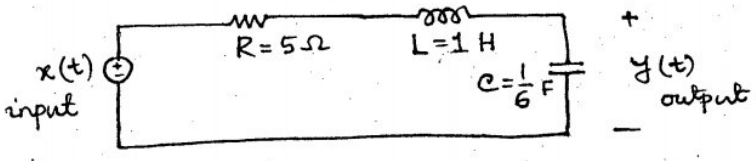


Fig. for Q. No. 5(b)

(c) By using the Cayley-Hamilton theorem, determine the state transition matrix $\Phi(t)$ from the state space coefficients obtained in part (b). **(10)**

(d) Considering that the initial state $\mathbf{V}_o$ is zero, find the impulse response of the system using the state-space coefficients and $\Phi(t)$ obtained in parts (b) and (c), respectively. **(8)**

6. (a) Is the following signal periodic? Justify. **(7)**

$$x(t) = exp(j\frac{7\pi}{6}t) + exp(j\frac{5\pi}{6}t)$$

If your answer is yes, find the fundamental period of x(t).

(b) Determine whether the following signal is energy or power signal or neither of these two types. **(7)**

$$x(t) = \begin{cases} 1, & t < 0 \\ e^{3t}, & t \geq 0 \end{cases}$$

(c) Write down the basic properties of Dirac delta function δ(t). Evaluate the following integral **(8)**

$$\int_{-\alpha}^{\alpha} e^{-5t+1}\delta'(t-5)dt$$

(d) For the function x(t) shown in Fig. for Q. No. 6(d) sketch y(t) = x(– 2t – 2) and write the analytical expression of y(t). **(13)**

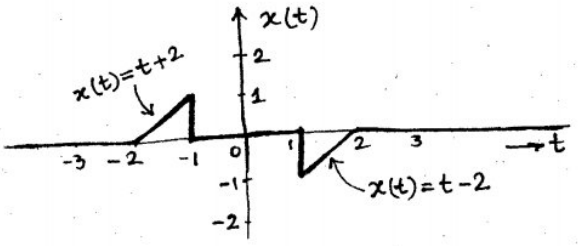


Fig. for Q. No. 6(d)

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7. (a) Determine whether the system shown in Fig. for Q. No. 7(a) is (i) linear, (ii) time-invariant, (iii) causal, and (iv) stable. **(8)**

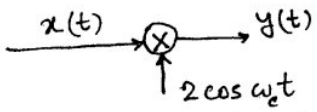


Fig. for Q. No. 7(a)

(b) Input output relationship of an LTI system is given by

$$y'''(t) + 3\,y''(t) - y'(t) - 2y(t) = 3x''(t) - x(t)$$ **(10)**

(i) Draw the simulation diagram for this LTI system based on canonical form-I.

(ii) How many adders, multipliers and integrators are used in your diagram?

(iii) Will it be possible to design an alternate simulation diagram for the same-system with reduced number of integrators?

(c) For an LTI system, output y(t) can be obtained from the convolution between input x(t) and the impulse response h(t). Mention the importance of this property in real life applications. **(4)**

(d) Consider the system defined in Fig. for Q. no. 7(d). **(13)**

(i) sketch g(t)

(ii) sketch y(t) and write the analytical expression of y(t). (Use graphical method of convolution integral.)

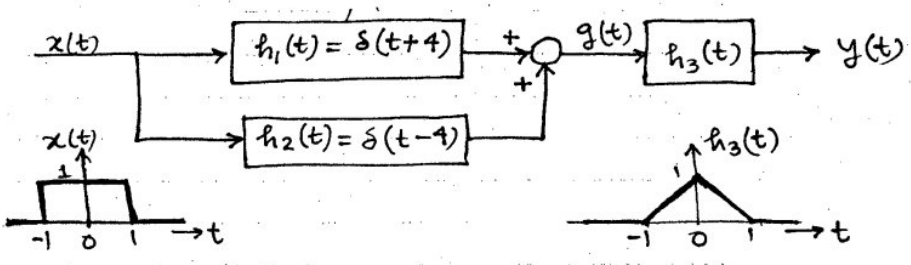


Fig. for Q. No. 7(d)

8. (a) Input output relationship of an LTI system is given by **(17)**

$$y''(t) + 5\,y'(t) + 6\,y(t) = x''(t) + 5\,x'(t) + 5\,x(t).$$

For an impulse input δ(t) to the system, following differential equation is obtained

$$h''(t) + 5\,h'(t) + 6\,h(t) = \delta''(t) + 5\,\delta'(t) + 5\,\delta(t).$$

Considering that the system is initially relaxed, determine h(t).

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Contd ... Q. No. 8

(b) Draw an analogous mechanical system for the electrical circuit shown in Fig. for Q. No. 8(b) **(8)**

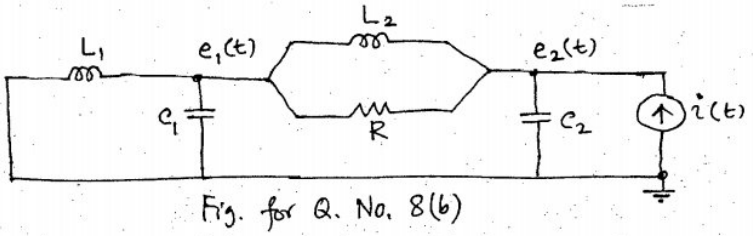


Fig. for Q. No. 8(b)

(c) Draw the f – i analogous electrical circuit of the mechanical system shown in Fig. for Q. No. 8(c). Assume that the bar is rigid but massless and that the ~~functions~~ junctions are restricted to have vertical motions only. **(10)**

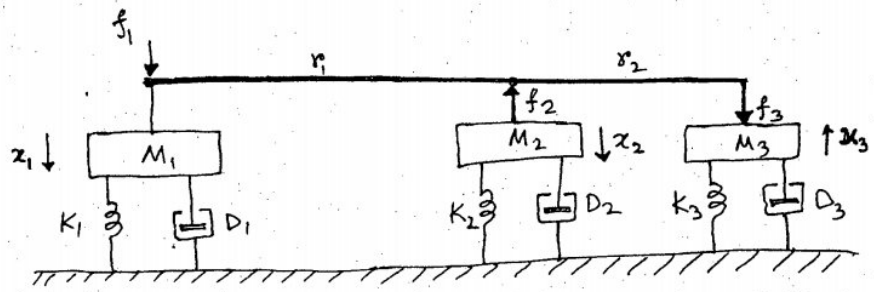


Fig. for Q. No. 8(c)

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L-3/T-1/EEE **Date : 19/03/2012**

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY, DHAKA

L-3/T-1 B. Sc. Engineering Examinations 2010-2011

Sub : **EEE 301** (Continuous Signals and Linear Systems)

Full Marks: 210 Time : 3 Hours

The figures in the margin indicate full marks.

Symbols bear their usual meanings.

USE SEPARATE SCRIPTS FOR EACH SECTION

SECTION – A

There are **FOUR** questions in this Section. Answer any **THREE**.

1. (a) The analytical representation of a square-wave with $x(t+T)=x(t)$ is given below: **(18)**

$$x(t)=\begin{cases}0, & -T/2<t<\dfrac{-\tau}{2}\\ K, & \dfrac{-\tau}{2}<t<\dfrac{\tau}{2}\\ 0, & \dfrac{\tau}{2}<t<\dfrac{T}{2}\end{cases}$$

(i) Draw $x(t)$ and find the Fourier Series Coefficients C_n of Fourier Series representation of $x(t)$

(ii) Sketch the magnitude spectra of $x(t)$ for $\{\tau=1, T=5\}$, $\{\tau=1, T=10\}$ and $\{\tau=1, T=15\}$. Comment on the effect of changing T on the frequency spectrum of $x(t)$.

(b) For an input $x(t)=A\exp\left|j10^3 t\right|$ applied to the RC circuit shown in Fig. for Q. 1(b) **(17)**

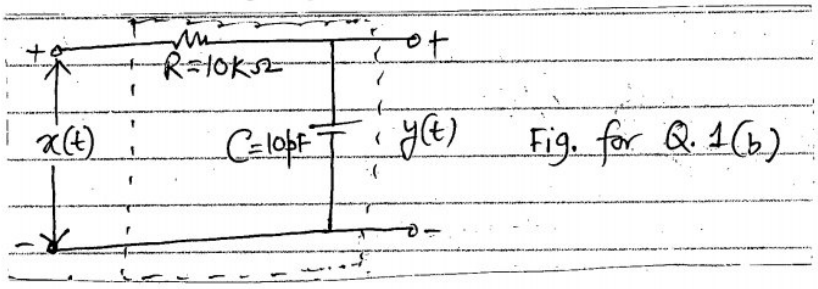


Fig. for Q. 1(b)

(i) Find the system function, magnitude and phase functions of the system.

(ii) Also, find the response $y(t)$.

2. (a) (i) Show that the Fourier Transform of a sequence of impulses in the time domain yields a sequence of impulses in the frequency domain. **(4×5=20)**

(ii) Show how the energy of an aperiodic signal can be computed in the Fourier domain.

(iii) Show that convolution in the time domain is equivalent to multiplication in the Fourier domain.

(iv) If a signal $x(t)$ is modulated by a cosine signal $\cos\omega_o t$, find the Fourier Transform of the modulated signal $x(t)\cos\omega_o t$.

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Contd ... Q. No. 2

(b) Consider the system in Fig. for Q. 2(b). **(15)**

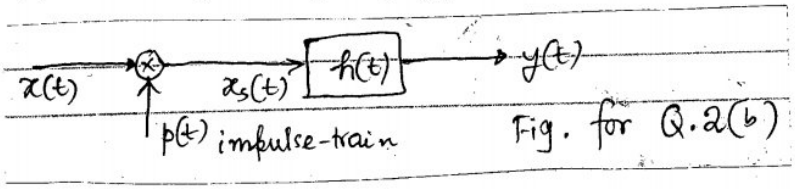


Fig. for Q.2(b)

where, $x(t)=\dfrac{\sin\left(\frac{\omega_B}{2}t\right)}{\pi t}$, $p(t)=\sum_{n=-\infty}^{\infty}\delta(t-nT)$.

(i) Sketch $X(\omega)$, the Fourier Transform Spectrum of $x(t)$
$P(\omega)$, the Fourier Transform Spectrum of $p(t)$ and
find the period of $p(t)$ from the expression of $p(\omega)$.

(ii) Find and sketch $X_s(\omega)$, the Fourier Transform spectrum of the product signal $x_s(t)$.

(iii) The system $h(t)$ blocks all the undesired components of $x_s(t)$ in order to obtain $y(t)$, which is a scaled version of $x(t)$. Write the expressions for $h(t)$ and $H(\omega)$. Sketch $H(\omega)$.

(iv) Also, find and sketch $Y(\omega)$.

3. (a) $Y(s)=L[y(t)]$ be the Laplace Transform of the solution $y(t)$ of the differential equation, $y''(t)+3y'(t)+2y(t)=0$, $y(0^-)=3$, $y'(0^-)=1$ **(12)**

(i) Find $Y(s)$

(ii) For the input $x(t)=\exp[-3t]u(t)$, find the transfer function H(s) of the system.

(iii) Find the final value of the response y(t).

(b) Using inverse Laplace Transform, find the impulse response of the system with transfer function $H(s)=\dfrac{s^2-3s+2}{s^3+6s^2+11s+6}$. Draw the two canonical form realizations for this system. **(6+17=23)**

4. (a) Consider the block diagram in Fig. for Q. 4(a). **(20)**

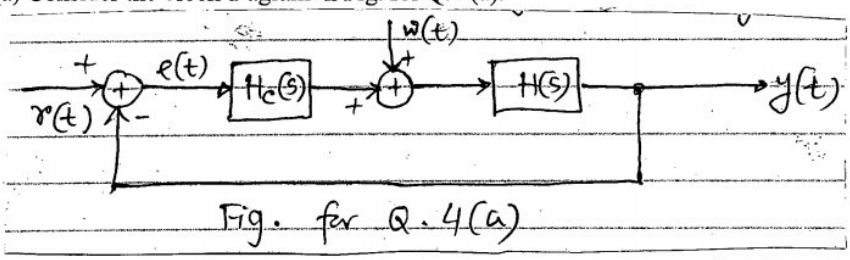


Fig. for Q. 4(a)

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Contd ... Q. No. 4(a)

input $r(t) = Au(t)$ and the disturbance $w(t) = Bu(t)$, where, A and B are constants.

(i) Find the Laplace Transform Y(s) of the output y(t)

(ii) Suppose $H(s) = \frac{N(s)}{D(s)}$ and $H_c(s) = \frac{N_c(s)}{D_c(s)}$. $H_c(s)$ has to be designed such that r(t) tracks y(t). Find the final value of y(t) with two necessary assumptions or requirements.

(b) State the D' Alembert's principle. Write the electrical and analogous mechanical differential equations for the electrical circuit shown in Fig. for Q. 4(b) using force-voltage (f-v) analogy. Also, draw the f-v analogous mechanical system for the circuit. **(15)**

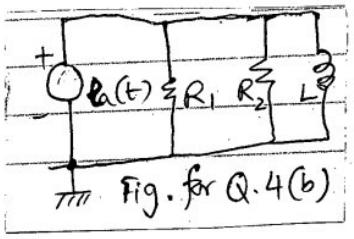


Fig. for Q.4(b)

SECTION – B

There are **FOUR** questions in this Section. Answer any **THREE**.

5. (a) (i) Is the signal $x(t) = \cos\left(\frac{\pi}{3}t\right) + \sin\left(\frac{\pi}{4}t\right)$ periodic? Justify and if periodic, determine the fundamental period of x(t). **(6+6=12)**

(ii) Consider the signal $x(t) = A\exp[-\alpha t]; t > 0$ and 0 otherwise. Find and sketch the odd and even parts of the signal.

(b) (i) Let $x(t) = u(t+2) - 2u(t+1) + 2u(t) - u(t-2) - 2u(t-3) + 2u(t-4)$. Sketch x(t). Find and sketch y(t) where $y(t) = \int_{-\infty}^{t} x(t)dt$. **(8+15=23)**

(ii) Sketch $x(t) = -t + 1; -1 \le t < 0$

$= t; \ 0 \le t < 2$

$= 2; \ 2 \le t < 3$

$= 0;$ otherwise

Sketch $x(t-2), x(t+3), x(-3t-2), x\left(\frac{2}{3}t + \frac{1}{2}\right)$ and find the analytical expressions for these functions.

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6. (a) Determine whether the following signals are power or energy signals. **(10)**

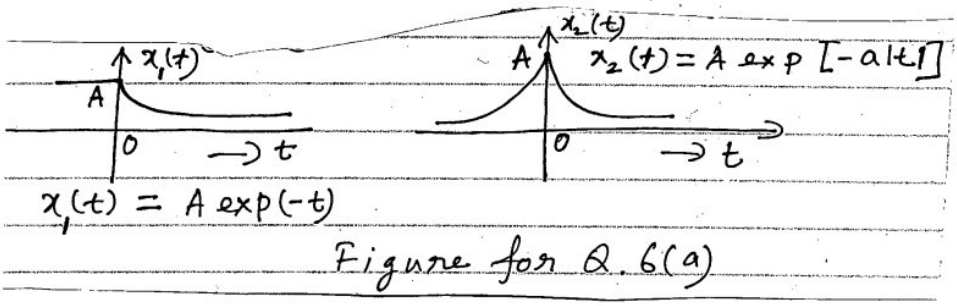


Figure for Q. 6(a)

(b) (i) The input and impulse response of a linear time invariant (LTI) system are denoted by x(t) and h(t), respectively, and are shown in the following figure. Using the graphical interpretation of convolution, obtain the output of the system, y(t). Also, write the analytical expression of y(t). **(10+15=25)**

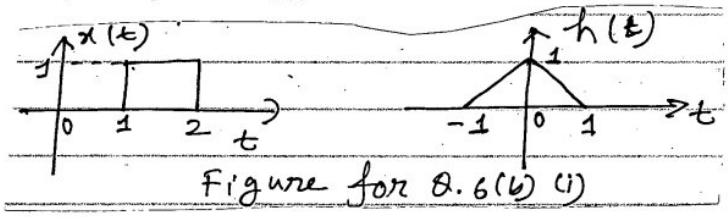


Figure for Q. 6(b) (i)

(ii) Find and sketch the impulse response of the LTI system shown below where $h_1(t) = \exp[-2t]u(t)$, $h_2(t) = \exp[-2t]u(t)$, $h_3(t) = \exp[-t]u(t)$, $h_4(t) = \delta(t)$ and $h_5(t) = \exp[-3t]u(t)$.

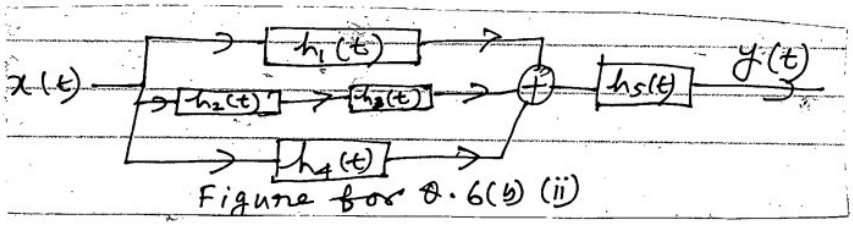


Figure for Q. 6(b) (ii)

Also, determine whether the system is causal and stable.

7. (a) The input-output relationship of a system is given by $y(t) = \mathrm{A}t\,x(t)$ where x(t) and y(t) represent the input and output, respectively. Determine whether the system is linear, time-invariant, causal and stable. **(10)**

(b) (i) Using state-variable techniques, find the impulse response for the system described by the differential equation: $y''(t) + 2y'(t) + 2y(t) = x''(t) + 3x'(t) + 3x(t)$ **(12+13=25)**

(ii) Find the state equations in the first and second canonical forms for the system described by the differential equation: $y''(t) + 2.5y'(t) + y(t) = x'(t) + x(t)$

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8. (a) The state-space model for an LTI system is given by **(10+8=18)**

$$v'(t)=\begin{bmatrix}0 & 1\\ 2 & -1\end{bmatrix}v(t)+\begin{bmatrix}0\\ 1\end{bmatrix}x(t)$$

$$y(t)=\begin{bmatrix}1 & 0\end{bmatrix}v(t)$$

(i) By using the Cayley-Hamilton theorem, comment on the stability of the system and determine the state transition matrix.

(ii) For a specified initial-condition vector

$$v_0=\begin{bmatrix}1\\ 0\end{bmatrix}$$

find the zero-input response and impulse response of the system.

(b) Consider the system described by the differential equation **(17)**

$$y''(t)+3y'(t)+2y(t)=x'(t)=x'(t)-x(t)$$

Find the transformation matrix P to convert the first canonical form into the second.

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